



Western Forest Products Limited

Tree Farm Licence 6
1998
Annual Report

Quatsino Tree Farm Licence 6 was granted to Western Forest Products' predecessor, British Columbia Pulp and Paper Company Limited, as Quatsino Forest Management Licence on October 25, 1950. On October 1, 1998, the Minister of Forests approved a significant change to the original Tree Farm Licence. Block 4 of TFL 25 was eliminated and amalgamated into TFL 6 as Block 2. This Block comprises the watersheds of Queen Charlotte Strait between Fort Rupert and Port McNeill, and portions of the Quatsino Sound watershed. Alaska Pine and Cellulose Limited, another predecessor company, was granted this Licence on May 22, 1958.

This Annual Report records accomplishments and data for the amended TFL 6 for the entire year except for detailed scale data. Scaled volumes reported reflect the amendment date of October 1, 1998. If you have comments on this report, please contact the Chief Forester at (604) 665-6224.

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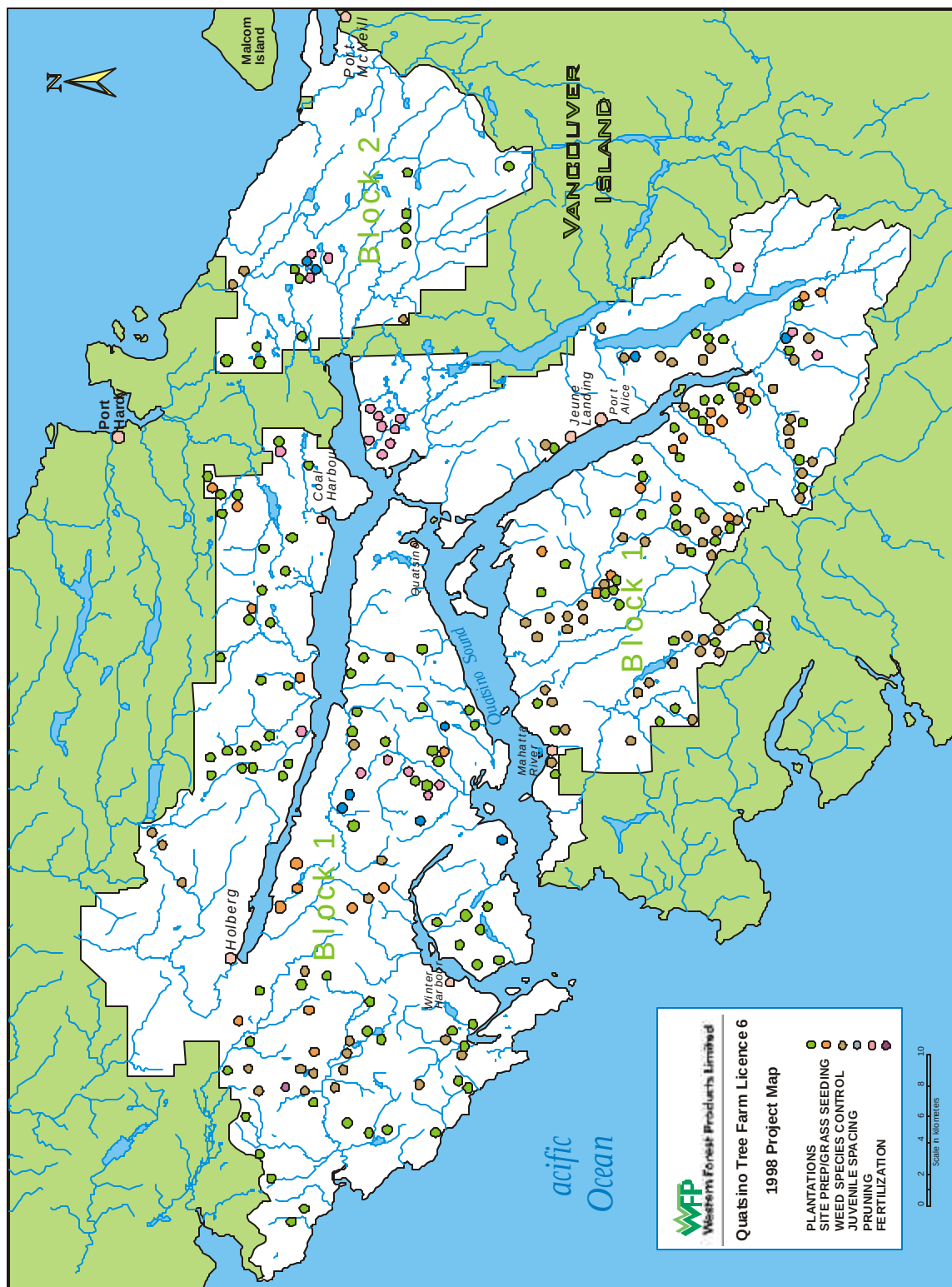
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SUMMARY OF ACTIVITIES AND ACCOMPLISHMENTS – 1998

Depletion	Scaled Volume	868 036.9 m ³
	Volume Charged to Annual Allowable Cut	909 007.8 m ³
	Area Logged (Western Forest Products)	1 104.7 ha
	Area Logged (SBFEP)	26.1 ha
Reforestation	Silviculture Prescriptions	2 200.8 ha
	Site Preparation	25.6 ha
	Seedlings Planted	1 473 595 trees
	Seedlings Fertilized	485 000 trees
	Area Planted	1 333.7 ha
	Stocking Surveys	2 548.3 ha
	Plantation Survival Assessments	1 444.1 ha
	Free Growing Surveys	11 304.5 ha
Stand Management	Juvenile Spacing	76.4 ha
	Brushing and Weeding	1 074.0 ha
	Pruning	256.6 ha
	Commercial Thinning	7.1 ha
Inventory	Cutting Permit Cruising	997 plots
	Residue Assessment	325 plots
Engineering	Roads Built	66.5 km
	Roads Re-Built	17.3 km
	Roads Maintained	767.9 km
	Roads Deactivated	53.4 km
	Roadside Seeding	99.1 km
	Site Stabilization	25.6 ha
Protection	Accidental Fires	3
Contracting	Contractor Obligation	335 531 m ³
	Contracted	470 710 m ³
	Compliance	140 %
Minor Products	Log Salvage	1 232.3 m ³
	Shake and Shingle	12 756.2 m ³
	Cants	63.9 m ³
	Yew Bark	2 932.0 kg
	Gravel and Sand	31 573.0 m ³
	Honey	18 135.0 kg
	Cedar Oil	512.0 litres
Salmon Enhancement Program (Fry Releases)	Chinook	211 900 fry
	Coho	164 056 fry
	Chum	20 600 fry
	Steelhead	3 200 fry
Employment	Direct Employment	174,177 person-days

PROJECT MAP



1.0 INTRODUCTION

1.1 Statement of Stewardship

Quatsino Tree Farm Licence 6 encompasses 201 500 ha of coastal forest on northern Vancouver Island. It has been managed by Western Forest Products Limited and its predecessors for 48 years. More than 60 million cubic metres of wood have been produced from these lands in that time. Forty percent of the total operable area of 157 000 ha remains in mature or old growth forest.

Internal and external audits and inspections indicated that a high standard of forest management was maintained during the year. Western Forest Products' Forest Operations were fully compliant with the Forest Practices Code and in many instances exceeded requirements.

Rural Vancouver Island communities, including Winter Harbour, Quatsino, Holberg, Port Hardy, Coal Harbour, Port Alice, and Port McNeill continued to receive significant economic benefits from the management activities on TFL 6. Despite extended shutdowns of harvesting operations in 1998, TFL 6 provided more than 935 direct full-time equivalent jobs in 1998. The economic and social benefits from TFL 6, combined with the Western Forest Products environmental stewardship program have resulted in a model of coastal forest management.

For more information on Western Forest Products Limited or TFL 6, visit the Company's web site at www.westernforest.com.



The view from Simpson 250 in the Holberg Operation looking west into Quatsino Sound entrance shows a mix of forest types. Approximately 50 per cent of the TFL forest cover remains in old growth condition, and healthy second growth covers the rest. Thirty per cent is not scheduled for harvest for environmental, social or economic reasons.

1.2 Operations Highlights

Western Forest Products' parent company, Doman Industries, had another difficult financial year in 1998. Year-end results were negatively affected by the depressed demand for hemlock lumber in Asian markets, continued low pulp prices, and high stumpage and logging costs. Losses for the year were \$74.6 million compared to a \$2.3 million loss in 1997. Sales for the year were \$779 million compared to \$737 million in 1997. The outlook for 1999 was not expected to be significantly better in spite of improved lumber and pulp prices at year end. The Company was successful in significantly reducing pulp and lumber inventories by year end. Poor markets resulted in the Company harvesting only 68 per cent of total Annual Allowable Cut in its forest operations in 1998.

Cost reduction and controlling an increasing bureaucratic and administrative burden was a major focus during the second half of the year. Government provided some stumpage relief in an attempt to keep logging operations active, but early in the fourth quarter all logging operations were shut down. Joint Industry-Government committees and action teams were put in place to design relief packages that would allow the forest sector to return to profitability in 1999. Appendix XIX summarizes the operating statistics for the Doman group of companies.



Yew logs pre-logged with a Hughes 500 helicopter had their bark stripped for taxol extraction. Note the timber mark that establishes "chain of custody" for forest certification.

Company Highlights

- On October 1, 1998, the consolidation of TFL 24 into TFL 25 and the addition of TFL 25 Block 4 to TFL 6 was approved by the Minister of Forests.
- Western Forest Products announced it was proceeding with Forest Certification to Forest Stewardship Council, International Standards Organization 14001, and Canadian Standards Association criteria.
- The Company signed a \$70 million Five-Year Agreement with Forest Renewal BC to finance a large watershed restoration, enhanced silviculture, and operational inventory program.

- Western Forest Products celebrated the planting of the Company's 75 millionth tree.
- The consolidation of the Lost Lake and Saanich Forestry Centre tree improvement orchards was continued in 1998. A number of orchards were removed and others enhanced. High breeding value hemlock clones were moved from Lost Lake and used to upgrade the Saanich orchards.

Specific TFL 6 Highlights

- At Holberg, the Quatsino Main #1 bridge was completed in the third quarter. The 46 m clear span accesses 1.5 million cubic metres of timber, and is designed to accommodate two loaded trucks simultaneously. The installed cost of the bridge was \$266,500.
- A permanent 27.5 m steel and concrete bridge was constructed across the Utluh River at Jeune Landing.

The new Quatsino Main bridge is a 46 m clear span that provides access to 1.5 million cubic metres of timber. The structure is comprised of three 15 m sections spliced and welded together. The installed cost for the bridge was \$266,500



- Jeune Landing continued operation of the Skagit 046 yarder acquired in 1997. Production has been better than expected. A total of 23 800 m³ were harvested with this longline machine.
- A total of 1986 hectares in the upper Mahatta River watershed was deleted from TFL 6 and added to the Kingcome TSA. This deletion was made as part of a volume exchange agreement between Doman Industries (FL A19237) and the Ministry of Forests in the Strathcona TSA. The area deletion will decrease the TFL 6 AAC by 12 494 m³.
- Commercial thinning trials continued at the Port McNeill Forest Operation resulting in further refinement of thinning methods. The feasibility of commercial thinning in coastal western hemlock forests remains questionable for both economic and forest health reasons.



Mr. Herb Doman, Chairman and President of Western Forest Products, and Roger Stanyer, Chairman of Forest Renewal BC, sign a multi-year Forest Renewal BC agreement. The \$70 million fund will focus on watershed restoration, enhanced silviculture, research, and operational inventory work.

- LeMare Lake Logging purchased Tribune Timber, and logged 120 000 m³ at Mahatta River. LeMare also logs for MacMillan Bloedel and Canadian Forest Products in the adjacent TSA area. This consolidation of contract operations with a single contractor will provide greater efficiencies for all companies.
- Shake and shingle block production at Holberg was 1 800 m³ higher than the previous year. Previously unsalvaged areas were temporarily accessible when the Watershed Restoration Program installed a 46 m portable bridge across the Goodspeed River. At Port McNeill, shake and shingle production was also high. A new operator began producing cedar oil from cedar foliage. Bough collection and processing employs a number of North Island residents.
- An agreement to exchange silviculture responsibilities was reached between the Ministry of Forests' Campbell River and Port McNeill Districts and Western Forest Products. Western Forest Products will assume responsibility for some of the Ministry of Forests Small Business areas in TFL 6 while the Ministry will assume silviculture responsibility for the Company's former Nootka Island operations in Forest Licence A19237.
- Holberg undertook a large free growing survey program in 1998. The program evaluated 3 115 ha of young, second-growth forests. Of the surveyed area, 82 percent was declared free growing. Minimal work is required on the remaining 18 per cent to achieve free growing status.
- The Forest Renewal BC enhanced forestry implementation crew was shut down in 1998. Reductions in Forest Renewal BC funding and rising unit costs as a result of New Forest Opportunities involvement in silviculture reduced the amount of work that needed to be prepared each year. Layout in most TFL 6 Operations is ahead of implementation.

- A herd of seventeen Roosevelt elk moved into the Goodspeed River watershed near Northeast 62 Road in Holberg operation. The herd includes three large bulls, eleven cows and three calves. One of the cows was radio collared from a previous Western Forest Products supported program.
- An Environmental Youth Team roughed out a new trail accessing West Coast beaches from Hecht Main near Winter Harbour.
- The O'Connell Lake Recreation Area was re-opened in 1998 in response to increased public demand. This recreation site offers day-use and overnight camping facilities and a small boat launch.
- Work continued on an extension to the Spruce Bay interpretive trail at Victoria Lake. The extension will provide examples of second growth management.
- The Cordy Creek Hatchery at Holberg made a record collection of brood stock with more than 440,000 coho, chum, and chinook eggs put into incubation.
- Work was started on the Marble River Rearing Channel. This major project is being built adjacent to Western Forest Products' Marble River campground and will draw water from the Marble River near the highway bridge. It will have the capacity to rear up to 2 million fry. The Friends of the Marble River raised more than \$250,000 in donations for this project.



As part of the Marble River Salmon Enhancement Project, the Friends of the Marble River completed Phase I of a new salmon-rearing channel in the Marble River Park. The channel will provide semi-natural rearing habitat for up to 2 million chinook fry with a gravity-fed water supply.

2.0 MANAGEMENT AND OBLIGATION PERFORMANCE

2.1 Volume

The scaled timber production for the year was 868 036.9 m³. Volumes by timber mark and Operation are summarized in Appendix I. The volume charged to the allowable cut including scaled production and residue survey volumes was 909 007.8 m³ (Appendix II). The TFL 6 volume for 1998 includes the prorated share of the former TFL 25 Block 4 volume from October 1, 1998 to December 31, 1998.



Western Forest Products' field engineer, Daniel Gachter, with old growth Douglas-fir tree near Michelson Point on Quatsino Sound. Veteran firs survived the hurricanes of 1904.

2.2 Area

The Company harvested timber from 1 104.7 ha during the year. Appendix III summarizes the denudation by Operation. Twelve per cent of the area logged was classified as medium, low, or very low ecosites. The low site partitioned AAC volume requirement was significantly exceeded (Appendix IX). The non-productive area in the TFL was increased by 66.5 hectares to account for the productive area occupied by new roads.

2.3 Contractor Compliance

Harvesting and forest management activities on TFL 6 during the year employed contractors and Company personnel. The minimum Contractor Clause commitment was exceeded by 40 per cent

(Appendix VI). Appendix VII lists the phase and full harvesting contractors. Appendix VIII lists all contractors employed in harvesting, silviculture, and resource management on the TFL.

2.4 Planning

2.4.1 Land and Resource Management Plans

During the year little progress was made by government to complete the Vancouver Island Land Use Plan process. Planning experts were drawn together by government to study and assess practices in Special Management Zones. Proposed development in the two TFL 6 Special Management Zones requires special studies and added assessments before harvesting can take place. The Company regularly reviews its practices in these areas.

Forestry costs associated with higher level planning, including land and resource management planning and landscape unit planning, were \$60,850.

2.4.2 Forest Development Planning

Forest Development Plans (FDP) were completed in all Operations according to Ministry of Forests' Policies and Procedures. The year saw the introduction of a revised Ministry FDP template that set new information and mapping standards. In order to improve the timeliness of FDP production and spread the administrative burden over more of the year, it was decided to split the two North Vancouver Island Regional Plans into four Forest Operation-based Plans. These were Holberg Regular, Holberg Special Management Zone, Port McNeill, and Jeune Landing. The preparation and production of FDP maps was also moved from Western Forest Products' corporate office to a Port McNeill mapping contractor. This required conversion of the Company's PAMAP GIS files to Microstation.

A one-year extension to the approved FDP was requested and received in order to provide time to introduce these new plans. Four new FDPs were submitted during the year but only the Jeune Landing Plan was approved in the same year.

Forest Development Plan preparation and compilation costs totalled \$143,966.

2.4.3 Silviculture and Harvest Planning

2.4.3.1 Cutting Permits and Cutting Permit Cruising

The number of active cutting permits in TFL 6 increased from 43 in 1997 to 51 in 1998 continuing the disturbing trend seen since the Ministry of Forests moved to Forest Licence styled permits for TFLs. Prior to this policy change only 9 roll-over cutting permits were required to harvest the same volume. Associated administrative costs have increased accordingly. Table 1 lists the active cutting permits, locations and expiry dates.

Cutting permit cruising plot costs decreased from 1997 costs. The lower average cost was due to a high proportion of plots in lower cost camps and the contractors' co-operation in a Company-wide cost reduction program. Average cost per established plot, Company-wide, was \$198. Average cost per compiled plot was \$68. The weighted plot cost was \$101. A total of 997 cutting permit cruising plots were established in TFL 6. Thirty cruise reports were compiled using data from 2,057 plots.

Total contractor cost was \$197,257. Forestry department costs associated with cutting permits and cutting permit cruising including contractor costs totalled \$274,172.

Table 1: Active Cutting Permits

CP	Expiry Date	Location	CP	Expiry Date	Location
100	Dec 31/98	Holberg	600	Mar 31/99	Coal Harbour
102	Mar 31/99	Holberg	601	Mar 31/00	Coal Harbour
103	Apr 30/00	Holberg	602	Feb 29/00	Coal Harbour
104	Jul 31/99	Holberg	7	Mar 31/98	Winter Harbour
2	Jan 31/98	Jeune Landing	700	Feb 29/99	Winter Harbour
200	Mar 31/99	Jeune Landing	701	Feb 29/00	Winter Harbour
201	Mar 31/99	Jeune Landing	702	Dec 31/99	Winter Harbour
202	May 31/99	Jeune Landing	703	Oct 31/00	Winter Harbour
203	Oct 31/99	Jeune Landing	750	Jul 31/99	Winter Harbour
204	Mar 31/00	Jeune Landing	751	Aug 31/00	Winter Harbour
205	Aug 31/00	Jeune Landing	752	Nov 30/00	Winter Harbour
300	Mar 31/99	Botel	800	Mar 31/98	Koprino
301	Jan 31/00	Botel	801	Feb 29/00	Koprino
302	Feb 29/00	Botel	802	Nov 30/98	Koprino
4	Jan 31/98	Mahatta River	803	Jan 31/00	Koprino
400	Mar 31/99	Mahatta River	804	Aug 31/99	Koprino
401	Feb 28/99	Mahatta River	870	Nov 30/98	Michelson
402	Aug 31/99	Mahatta River	871	Dec 31/98	Michelson
403	Sep 30/00	Mahatta River	872	Dec 31/99	Michelson
4M	Aug 31/99	Port McNeill	873	Jul. 31/00	Michelson
40M	Jan 01/00	Port McNeill	900	Jan 31/00	Rupert Inlet
42M	Sep 30/99	Port McNeill	901	Jan 31/00	Rupert Inlet
43M	Oct 31/99	Port McNeill	902	May 31/00	Rupert Inlet
44M	Dec 31/99	Port McNeill	90	Oct. 31/99	All Areas
500	Nov 30/00	Port McNeill	98	Apr 14/99	Various
61	Feb 28/99	Coal Harbour			

2.4.3.2 Road Permits

The Company maintained three Road Permits in 1998, as listed in Table 2.

Table 2: Active Road Permits

Road Permit	Expiry Date	Operation
R06936	Open	Port McNeill
R06897	Open	Jeune Landing and Mahatta River
R06896	Open	Holberg, Botel, and Winter Harbour

2.4.3.3 Silviculture Prescriptions

Sixty-five new silviculture prescriptions were submitted during the year (Table 3). The total area for which new silviculture prescriptions were submitted was 2 200.8 ha. Average block size including reserves was 33.9 ha. Average reserve area varied from 10.7 per cent to 15.1 per cent by Operation, with a weighted average of 13.9 per cent.

Amendments to silviculture prescriptions continue to occupy significant staff time. Forestry costs for new silviculture prescriptions and amendments to previously submitted prescriptions totalled \$168,925 or \$77 per ha.

Table 3: Silvicultural Prescriptions

Operation	Submitted SPs (#)	Total Area (ha)	Average Block Size (ha)	Minimum Block Size (ha)	Maximum Block Size (ha)	Average Reserve (%)
Holberg	31	1034.4	33.4	5.1	55.4	14.5
Jeune Landing	16	468.6	29.0	9.0	46.1	10.7
Port McNeill	18	697.8	38.7	2.7	47.2	15.1
TOTAL	65	2200.8	33.9	2.7	55.4	13.9

Paul Watson, R.P.F., of the BC Assessment Authority, confirms success of a 1990 western red cedar plantation within Managed Forest 31. TFL 6 includes large tracts of private land that have been committed to sustainable forest management.



2.5 Public Involvement

2.5.1 Forest Development Plans

During the year, two TFL Forest Development Plans (FDP) were replaced by four new submissions. The four new plans for Port McNeill, Jeune Landing, and Holberg (2) cover the areas in the vicinity of each Forest Operation. Administratively Holberg has two FDPs because of the Special Management Zones within their operating area. Public response to all four draft Plans was solicited

through media notification and open houses. A summary of the responses from the public is detailed in Table 4.

Special presentations were also made to the Kwakiutl and the Quatsino First Nations to explain activities within their traditional territories.

Forestry Costs to arrange FDP public viewings totalled \$6,433.

Table 4: Forest Development Plan Reviews

Operation	Plan	Location	Attendance	Written Responses	Oral Responses
Jeune Landing	Jeune Landing FDP	Port Alice	17	2	
Port McNeill	Port McNeill FDP	Port McNeill	14	2	4
Holberg	Holberg FDP	Holberg	13	0	
		Port Hardy	12	1	
		Winter Harbour	15	10	
	Holberg SMZ FDP	Holberg	5	0	
		Port Hardy	20	0	
		Winter Harbour	15	10	
TOTAL	4	8	111	25	4

2.5.2 Forest Education and Special Events

The Company continued its support of the North Island Forestry Centre (NIFC). This non-profit society co-ordinated summer tours of logging operations on the North Island. The Port McNeill Forest Operation hosted a total of 374 guests on twenty-two tours. These tours were able to observe forest management activities and discuss forest practices with Western Forest Products employees. Four other tours visited the Beaver Lake interpretive trail and learned about coastal forests and the changes initiated by natural disturbance or harvesting. A local contractor was hired to host the 1998 tours.

All Operations hosted or spoke to school groups. Jeune Landing Operation discussed forestry careers with Port Alice students on Careers Day. Holberg Foresters made presentations on forests and forest management to 560 students in preparation for National Forest Week. Other students toured the Holberg Forest Operation and hatchery. Port McNeill Operation hosted 5 local student/teacher tours as well as a special tour for fifty-five exchange students. Presentations were also made to eighty students at Coal Harbour.

Port McNeill Operation and the North Vancouver Island Region hosted numerous special tours. A group of 22 Swedish Foresters visited North Island Operations as part of a BC tour. The Rockefeller Brothers Fund Incorporated sent two delegates to examine forestry practices. Science

Council representatives, MacMillan Bloedel staff, and a group of European geneticists toured Western Forest Products research trials established in the Port McNeill area.

Western Forest Products' \$70 million Multiyear Agreement with Forest Renewal BC was signed at a special ceremony at the Rupert Arm Recreation Site. Company representatives, including Mr. Herb Doman (Chairman and President of Western Forest Products), Mr. Vic Woods (Vice President and General Manager), and Mr. Bill Dumont (Chief Forester), together with Forest Renewal BC representatives, including Roger Stanyer (Chairman of Forest Renewal BC), and Ron Bronstein (Regional Director of Pacific Region), welcomed community and labour groups to the signing ceremony.

The Company spent \$72,747 on tours and special events in TFL 6. In addition, the Company's Public Projects fund contributed \$17,580 to Forest Education efforts in the TFL. Public relations costs associated with Forest Renewal BC events or projects are covered by the Multi-year Agreement.

2.6 Inventories and Mapping

2.6.1 Geographic Information System and Electronic Mapping

Five PAMAP GIS stations were used for corporate resource inventory mapping and analysis. The purchase of ten PAMAP Viewer licences and increased use of Microstation to produce Forest Development Plans enabled increased use of digital map data throughout the Company. These two mapping formats allow data exchange.

Dendron Resource Surveys Inc. prepared external dBase entry application software to enable use of expanded databases associated with new vegetation resources inventory.

Training opportunities provided by PCI-Pacific and BCIT were used to enhance the skills of GIS personnel.

Prorated TFL 6 GIS costs totaled \$84,188.

2.6.2 Forest Inventory

TFL 6 digital files were updated to January 1 1997, to incorporate changes to forest cover, roads, operability and logging history that occurred during 1996. Construction of a logging history digital level was completed. Further update of the digital files to January 1 1998 was started, to permit forthcoming timber supply analyses to be based on this more recent data.

With assistance from Dendron Resource Surveys Inc. digital capture of Forest Practices Code stream classification available to date was completed for both Blocks 1 and 2. The addition of newly contoured area and fill-in of 10-metre contours by Dendron continued with the work focused on Jeune Landing, Coal Harbour, and Drake Island. Operational mapping at 1:5 000 scale was

completed by Kjell Hoel for 8 520 ha at Drake Island, Holberg, Port McNeill, Victoria Lake and Winter Harbour. In-house capture of revised operability classification for the TFL was begun.

Database, map, and aerial photography information was provided to the Ministry of Forests to assist their organization of a TFL 6 inventory audit.

TFL digital mapping control was translated from the current polyconic NAD27 datum to the BCGS UTM NAD83 datum, as a preliminary to a NAD shift to be undertaken at a later date.

Costs for forest inventory work totaled \$168,559.

2.6.3 Ecosystem

In conjunction with Block 2 terrain stability revision, some fine-tuning of the ecosystem classification was made by Dr. Terence Lewis and incorporated into the GIS. Costs for ecosystem work totaled \$14,911.

2.6.4 Terrain Stability

The revision of the 1992 Block 2 terrain stability classification to Ministry of Forests standards was completed by Dr. Terence Lewis and entered into the GIS. The Block 1 terrain stability, completed by Dr. Lewis in 1995, already met Ministry of Forests standards. Costs for terrain stability work totaled \$5,820.

2.6.5 Geotechnical Analyses

Engineering consultants were employed to complete geotechnical analyses of cutblocks and road construction and deactivation projects throughout TFL 6. Table 5 details the consultants employed by Operation. Costs for geotechnical analyses are included in logging costs.

Table 5: Geotechnical Analyses

Operation	Geotechnical Firm	Areas	Roads
Jeune Landing	Terra Firma	7	2
	Thurber	4	1
	Westrek	6	1
Port McNeill	Arbor Tech	6	
	Terra Tech	2	
Holberg	Terra Engineering Group	20	1
	Bruce Geotechnical Consultants Ltd.		2
TOTAL		45	7

2.6.6 Resource Inventories and Assessments

A large number of inventories and assessments were completed during the year, many to support the preparation of operation plans. Except where indicated, costs for these inventories and assessments were incurred by the Forestry Department, and totalled \$198,682.

2.6.6.1 Archaeology

The Holberg Operation employed Sources Archaeological & Heritage Consultants to complete 11 Archaeological Impact Assessments of cut blocks in the Holberg, Winter Harbour, and Koprino operating areas. Representatives of the Quatsino First Nation were involved in all of these assessments. Operational plans were adjusted to protect Culturally Modified Trees and other archaeological features identified in these assessments. Archaeological Impact Assessment reports were provided to the Quatsino First Nation, the Ministry of Forests and the Heritage Conservation Branch. Costs for cutblock archaeological assessments are included in logging costs.

Two impressive culturally modified trees (CMTs) along the Kwatleo River in Winter Harbour form part of the heritage and evidence of occupancy by the Quatsino First Nations people. Policies are evolving regarding the protection and conservation of culturally modified trees. Western Forest Products identifies, maps, and informs government and First Nations of these assets.



2.6.6.2 Streams

Stream classification assessments were carried out on 39 cutblocks and roads during the year in advance of final cutblock and road engineering. This information was used to designate riparian reserves and riparian management zones in each cutblock. Fishfor Contracting completed all surveys. Costs of stream classifications are included in logging costs.

Western Forest Products also began the first phase of a Special Management Zone fisheries inventory program. The 1998 project was done to Resource Inventory Committee (RIC) standards. While the Company disagreed with some of the RIC methodology, the Company agreed to use these standards for the first year to determine baseline costs.

Forestry department costs for stream assessments and inventory work totalled \$80,036. Reimbursements from Forest Renewal BC for landscape level work were \$68,893.

Fisheries contractors undertake electrofishing on Kewquodie River Drainage in Mahatta River as part of forest development planning. Electrofishing is effective in determining fish presence or absence. Stream classification assessments and detailed riparian management prescriptions are required under the Forest Practices Code.



2.6.6.3 Wildlife

Wildlife work in TFL 6 has historically focused on the enumeration of individual species. This approach is very expensive and does not address broader habitat or ecosystem issues. In 1998 Western Forest Products approached the University of British Columbia Centre for Conservation Biology to help develop a planning model that would allow the identification of high value habitat for sensitive species on a landscape basis. A project to test the effectiveness of this approach was prepared for implementation in early 1999. Forestry department costs were \$78,216. Reimbursements from Forest Renewal BC totalled \$38,427.

2.6.6.4 Recreation

The TFL 6 Recreation inventory was reviewed during the year. As there have been no significant changes in recreational use of TFL lands it was not considered necessary to revisit the inventory. Forestry department costs totalled \$12,068.

2.6.6.5 Landscape Assessments

The Ministry of Forests approached all District Licensees in mid-year to review visual management issues with the intention of mitigating impacts on District timber supplies. Western Forest Products staff helped to identify key visual areas and opportunities to minimize economic impacts. Final comments were submitted at the end of the year. Department costs totalled \$12,873.

3.0 MANAGEMENT OBJECTIVE ACHIEVEMENTS

3.1 Management and Utilization of Timber Resources

3.1.1 Harvesting Methods

Five logging systems were used in TFL 6 in 1998 (Figure 1). In areas managed by Jeune Landing (including Mahatta River) grapple yarders harvested the largest portion of the volume (39 per cent) followed by the supersnorkel (23 per cent) and the tower (21 per cent). Both hoe forwarding and skyline logging systems accounted for 8 per cent of the volume harvested at Jeune Landing.

Holberg, Botel, and Winter Harbour used grapple, super snorkel, and hoe forwarding logging systems (Figure 1). Grapple yarders harvested 44 per cent of the volume, super snorkels 32 per cent, and hoe forwarders 24 per cent.

Port McNeill Forest Operation used hoe forwarding, a tower, a super snorkel, and a grapple (Figure 1). Almost sixty per cent of the volume was handled by super snorkel. Grapple yarding accounted for 18 per cent of the volume, and hoe forwarding accounted for 10 per cent of the volume. The entire tower logging (13 per cent) was completed by Quatsino Forestry.

Port McNeill Operation also continued with commercial thinning trials. The last system utilized was a small excavator, which forwarded material to skid trails. All systems used over the past three years (cable, cut-to-length forwarding, and hoe-forwarding) have been disappointing from both an economic and a forest health point of view. Forestry department costs associated with this trial were \$4,500 in 1998.

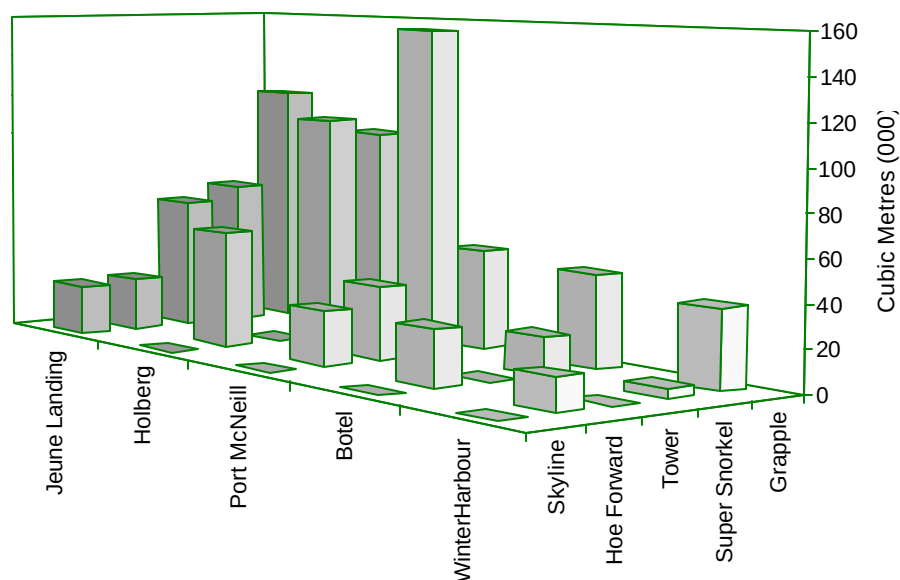


Figure 1. Logging Volumes by Harvest System and Operation.



Jeune Landing Forest Operation experimented with a mini skyline system in cut block 622 in May of the year. The lightweight skyline uses a radio-controlled self-propelled carriage. Front end lift is provided by a backhoe on a road. The system may be a low-cost option for partial cutting or salvage in terrain with suitable deflection.

3.1.2 Silviculture Systems

Clearcut with reserves is the main silvicultural system employed at all Operations. Thirty-three of the 40 areas harvested in 1998 were clearcut with reserves. Fifty-four of the 59 blocks engineered during the year will use this system. Average reserve area exceeds ten per cent of the gross area of each block. Over one-fifth of the cublocks in operational plans currently meet non-clearcut retention criteria. Table 6 provides a silviculture system summary by Operation.

Table 6: Silvicultural Systems by Operation

Operation	System	Blocks Engineered (#)	Average Block Size (ha)	Minimum Block Size (ha)	Maximum Block Size (ha)
Holberg	Clearcut with Reserve	28	34.7	5.1	49.2
	Clearcut	1	6.3	6.3	6.3
Jeune Landing	Clearcut with Reserve	19	31.8	9.0	44.2
Port McNeill	Clearcut with Reserve	11	34.2	7.3	45.8
Operation	System	Blocks Harvested (#)	Average Block Size (ha)	Minimum Block Size (ha)	Maximum Block Size (ha)
Holberg	Clearcut with Reserve	18	34.5	14.1	50.2
	Clearcut	1	34.2	na	na
Jeune Landing	Clearcut with Reserve	2	37.6	29.1	46.0
	Clearcut	5	31.9	17.3	39.4
Port McNeill	Clearcut with Reserve	13	30.1	8.2	41.6
	Commercial Thinning	1	7.1	na	na

Commercial thinning trials continued in TFL 6, Block 2, near Port McNeill, during the year. While the "hoe-sliding" trial was not commercially successful, new experience in alternate harvesting techniques was gained and approximately 3 300 cubic metres in thinnings were harvested.



3.1.3 Felling, Bucking, and Utilization Specifications

3.1.3.1 Specifications

Cutting permits outline the required and optional utilization standards. In general:

- i. Maximum stump height of 30 cm on shortest side;
- ii. All old growth coniferous trees containing X grade logs or better will be utilized to a top diameter of 15 cm inside bark (10 cm for second growth) ;
- iii. All conifer trees or parts of these trees exceeding 3 m in length which contain X grade logs or better will be removed;
- iv. Logs or parts of logs less than 3 m in length and broken at ends are classed as breakage;
- v. Logs will not be bucked or trimmed in a manner that reduces grade.

These obligations apply to all living and dead trees that meet the standards. Quality control is maintained by Western Forest Products' staff and monitored through felled and bucked inspections and residue surveys.

3.1.3.2 Residue Surveys

The Company continues to survey and report residue volumes on a calendar year basis for all cutblocks where logging was completed during the year. Market induced production curtailments significantly reduced the area surveyed in 1998. A total of 325 plots were established.

The official Ministry of Forests residue volume used for the Annual Allowable Cut depletion is determined by summing the volumes from the Scale and Royalty invoices billed during the calendar year. As a consequence, 1998 depletion includes residue volumes from 1997 and 1998 surveys as

determined by the billing date. For 1998, the billable waste totalled 9 294.2 m³ and the Annual Allowable Cut depletion totalled 29 474.6 m³ (Table 7).

The contractor costs for residue sampling and compilation were \$34,574 or \$106.38 per plot. Total costs of residue assessments were \$75,817.

Table 7: Residue Assessment Survey Summary

LOCATION	SURVEY AREA		AAC DEPLETION		BILLABLE WASTE TOTAL VOLUME		NUMBER OF PLOTS				
	Gross (ha)	Net* (ha)	m ³ /ha	m ³	m ³ /ha	m ³	Slash	Road Side	Pile & RMZs	Other	TOTAL
Holberg	38.9	37.5	105.4	3954.1	13.3	497.8	10	4	6	2	22
Winter Harbour	94.6	91.0	64.0	5822.2	15.3	1388.3	28	16	20	4	68
Koprino	155.1	147.9	45.3	6706.2	11.4	1684.5	30	24	30	6	90
Botel Lake	75.0	72.3	29.9	2158.9	7.2	522.5	21	8	12	2	43
Coal Harbour	97.5	87.7	26.3	2305.8	8.2	720.5	12	12	12	2	38
Jeune Landing	138.0	133.3	64.0	8527.4	33.6	4480.6	18	16	26	4	64
TOTAL	599.1	569.7	51.7	29474.6	16.3	9294.2	119	80	106	20	325

3.1.3.3 Special Forest Products

Special Forest Products harvested in 1998 included salvage logs, cedar shake and shingle blocks, cedar cants, Sitka spruce guitar blocks, yew bark, honey, cedar oil, and gravel (Appendix XI). Salal was harvested in the TFL; however, amounts were not reported.

More than 12,000 m³ of shake and shingle blocks were salvaged (Appendix XI). Jeune Landing reported similar amounts to the previous year's salvage at 2 164 m³. Port McNeill reported lower (15 per cent) salvage volumes for shake and shingle than the previous year, at 5 259.5 m³. Holberg reported salvage of 5 332.7 m³, up 30 per cent from the previous year's production.

Salvage of logs occurred at Jeune Landing and Port McNeill (Appendix XI). These included cedar and yew logs. At Holberg, cedar cants were salvaged for processing at local Port Hardy sawmills. The volume of cants was down significantly from recent years.

Yew bark harvesting continued at Port McNeill and Jeune Landing. Total weight of the 1998 harvest was 2 932 kg (Appendix XI), up 65 per cent from the previous year's harvest.

Port McNeill Forest Operation sold 31 573 m³ of gravel in 1998. This represents a 42 per cent increase from the previous year.

A new contractor began producing cedar oil at Port McNeill Operation. Processing of cedar boughs produced 512 litres of cedar oil. Honey production was down at 18 135 kg (Appendix XI).

The Forestry Department administered the Minor Products program at a cost of \$43,169.

Susan Desbiens of Blue Canyon Enterprises examines yew logs. Yew bark and wood are examples of minor forest products creating opportunities for small business ventures. In this case, bark has been stripped for taxol extraction and logs have been flown out by helicopter prior to general timber harvesting.



3.1.4 Partitioned Harvest

The Annual Allowable Cut for TFL 6 has been partitioned into Conventional, Helicopter, and Low Site Volumes. Appendix IX reports the 1998 partitioned volume as required in Management Plan No. 9. The low site volume is estimated by matching low site harvest area to inventory volumes. On this basis Western Forest Products' produced 122 476 m³ from low site lands which significantly exceeds the low site target of 52,000 m³ per year. The Company did not undertake any helicopter logging in TFL 6 during the year due to depressed markets so none of the 10,200 m³ of partitioned helicopter wood was harvested during the year.

3.1.5 Annual Allowable Cut and Cut Control Performance

The Allowable Cut available to Western Forest Products on TFL 6 for 1998, including the prorated Allowable Cut available from the former TFL 25 Block 4, was 1 323 084 m³ (Appendix IV). The chargeable cut for the year, including the scaled production and the recognized residue survey volumes, was 909 008 m³ (Appendix II). Chargeable volumes for the former TFL 25 Block 4 were prorated from October 1. The cumulative volume charged after four years of the current five-year cut control period is 4 828 852 m³ or 74 per cent of the five-year total allowable cut (Appendix IV).

The historic cut control is presented in Appendix V. Western Forest Products and its predecessors have harvested 48 585 372 m³ since the granting of the Licence in 1950. To the end of 1998, the accumulated Allowable Cut was 48 519 266 m³. The rate of harvest is at 100.1 per cent of the Allowable Cut to date as determined by the Chief Forester of BC.

3.1.6 Small Business Forest Enterprise Program

The Small Business Forest Enterprise Program (SBFEP) has been in place since 1988. There has not been any new Block 1 SBFEP volume added to the program since a 1996 volume exchange agreement was reached with the Ministry of Forests. At the time that the agreement was made, there was still a "backlog" of TFL 6 SBFEP volume that the Ministry had not issued as Timber Sales. This backlog volume is still available to the Program. However, none of this backlog volume was harvested in 1998.

The SBFEP allocation in Block 2 (the former TFL 25 Block 4) is 13 242 m³ per year. The Port McNeill Forest District reported that 25.3 ha of Block 2 was harvested by B.W. Creative Wood Industries under the SBFEP. The chargeable volume was 22 526 m³ (Appendix X).

3.2 Protection and Conservation Measures

Western Forest Products protected and conserved identified resource values in TFL 6 during the year. The following sections review planning and operational measures taken to protect other resource values. Blocks for which amendments were submitted and blocks in which harvesting was not completed are not included in Table 8.

3.2.1 Visual Quality

Four of the 59 blocks engineered during the year required special engineering to address visual concerns (Table 8). Five of the 39 blocks harvested during the year had been similarly modified to address visual issues. Modification includes movement of backlines and the location of reserves to minimize the visual impacts of harvesting. Areas where visual quality concerns exist were identified in Forest Development Plans.

3.2.2 Biological Diversity

Biological diversity issues were addressed through the establishment of stand level reserves as well as landscape level deferrals. In addition to Code reserves, Holberg reported special conservation measures were taken in 16 of 29 engineered blocks and 6 of 19 harvested blocks (Table 8). These special measures were addressed in the Silviculture Prescriptions.

Table 8: Protection and Conservation Measures.

Operation	Blocks Engineered	Number of Blocks Requiring Special Consideration							
		Visual Quality	Biological Diversity	Soil Stability	Water Quality	Cultural Heritage	Fish Habitat	Wildlife Habitat	Wind Firming
Holberg	29		16	13		1	13		
Jeune Landing	19	4		12	2		3		
Port McNeill	11			7			10		
TOTAL	59	4	16	32	2	1	26		
Operation	Blocks Harvested	Number of blocks Requiring Special Consideration							
		Visual Quality	Biological Diversity	Soil Stability	Water Quality	Cultural Heritage	Fish Habitat	Wildlife Habitat	Wind Firming
Holberg	19	3	6	7	1	1	15		1
Jeune Landing	7	1						1	
Port McNeill	13	1		3			12		
TOTAL	39	5	6	10	1	1	27	1	1

3.2.3 Soil Stability Maintenance

3.2.3.1 Conservation

Soil resources were protected at all Operations (Table 8). Ten of 39 harvested blocks required special stability assessments. Thirty-two of 59 engineered blocks required special stability assessments.

3.2.3.2 Operational Site Stabilization Program

Erosion control treatments were carried out at Jeune Landing (7.5 ha), Holberg (14.6 ha), and Port McNeill (3.5 ha) Operations (Appendix XI). Grass seeding of cut slopes, slides, and sensitive sites in current logging blocks is a Company responsibility under the Forest Practices Code. Costs for the site stabilization program totalled \$70,615.

3.2.3.3 Watershed Restoration Program

Forest Renewal BC fully funded ten watershed restoration projects in TFL 6 (Table 9). French Creek Forest Services Ltd co-ordinated the program on behalf of the Company. Accomplishments included 39.8 km of road deactivation, 8.2 km of fish habitat restoration, and 7.2 ha of hydro seeding. Program implementation and administration costs were \$1,805,302.

Table 9: Watershed Restoration Program Summary

Watershed	Level I	Level II	Road Deactivation	Fish Habitat Restoration	Up-slope Restoration	Spending
Mahatta				1.2 km		\$243,028
Goodspeed			14.2 km			\$697,923
Hushamu		⊗				\$11,215
Johnson		⊗	24.9 km			\$383,647
Keogh		⊗		5.0 km		\$145,063
Marble	In Progress	In Progress				\$95,384
San Josef			0.7 km	2.0 km		\$138,841
SE 390						\$65,285
Jeune Landing Hydro Seeding					7.2 ha	\$16,905
North Holberg Inlet					In Progress	\$8,011
TOTAL	1	4	39.8 km	8.2 km	7.2 ha	\$1,805,302

3.2.4 Water Quality

Special engineering to maintain water quality was required for 2 engineered blocks at Jeune Landing and 1 harvested block at Holberg (Table 8).

3.2.5 Recreation Resources

3.2.5.1 Public Recreation Sites

Maintenance of public recreation sites, including provision of firewood and cleaning of facilities was carried out during the summer tourist season. At Jeune Landing, Forest Renewal BC funded the first phase of the Spruce Bay trail extension. The same crew completed an extension of the Beaver Lake trail during the year.

The O'Connell Lake Recreation Area, in the Mahatta watershed, was re-opened in 1998. This facility allows for overnight camping and small boat launching.

An Environmental Youth Team crew sponsored by Western Forest Products completed numerous TFL 6 trail and recreation site upgrade projects. These included work on Hecht Main, Bear Creek, Varney Bay, Brown's Bridge, and Cluxewe trails. The crew brushed out the Palmerston, Koprino, Swan Lake, and Teeta Lookout Recreation Sites.

The Company's Public Projects fund covered \$27,371 of the recreation site maintenance and upgrading costs. The Forestry Department spent an additional \$47,914 on Western Forest Products and Forest Renewal BC recreation projects. Forest Renewal BC funds reimbursed \$32,721 of these departmental costs.

3.2.5.2 Recreation Use

There is extensive use of TFL 6 for dispersed recreation. Estimates of annual use are presented in Table 10. These estimates are based on visits recorded to maintained sites, issued permits, and information from the tourism industry. An unusually dry summer resulted in heavy site use. Saltwater fishing numbers were depressed by the coast wide coho salmon fishing closure.

Table 10: Recreation Use Estimates

Activity	User Days By Operation			
	Holberg	Jeune Landing	Port McNeill	Total
Beach Use	1600	100	850	2550
WFP Sites and Trails	3400	4000	6500	13900
Hunting	350	800	750	1900
Fishing (Freshwater)	350	2000	1200	3550
Fishing (Saltwater)	1000	2000	1500	4500
Firewood Cutting	250	1000	680	1930
Food Gathering	100	500	250	850
Kayaking	450	300	75	825
Whale Watching	50	0	0	50
Auto Touring	9500	4000	15000	28500
Hiking and Caving	750	200	200	1150
TOTAL	17800	14900	27005	59705

3.2.6 Cultural Heritage Resources

All cultural heritage values identified through inventories and assessments are protected during development. When undocumented resources are discovered, operational plans are altered to ensure their protection. Port McNeill Operation modified plans for one future block for cultural heritage values in 1998 (Table 8). Holberg Forest Operation modified harvest plans in an active block when fallers discovered a Culturally Modified Tree (CMT).

A gate was installed on Suquash Main in Block 2 to limit vehicle access to the Suquash heritage site. This gate was requested by both the Regional District of Mount Waddington and the Kwakiutl Band Council to protect the First Nations and European heritage values in this area.



A surveyor with Sources Archaeological & Heritage Consultants assesses this culturally modified tree in the Botel Operation.

3.2.7 Fish Habitat

3.2.7.1 Conservation

Protection and conservation of fish and fish habitat requires the designation of riparian reserve and riparian management zones at the engineering stage. In 1998, 26 of 59 engineered blocks required special management to protect fisheries values. Twenty-seven of 39 harvested blocks had required similar attention (Table 8). Harvesting requires special falling and yarding techniques near these zones as well as specialized treatments such as aerial pruning to increase windfirmness.

3.2.7.2 Salmonid Enhancement Program

Western Forest Products Limited continues to support Salmonid Enhancement projects in its TFLs. Local volunteers, many of whom are Company personnel, provide volunteer labour and skills to these projects. Hatcheries and habitat improvement projects continue to be a highlight of our forest tours.

TFL 6 watersheds enhanced through the Company's support of salmonid enhancement projects include the Goodspeed and San Josef Rivers near Holberg, the Marble River south of Port McNeill, and the Colonial and Cayegle Creeks south of Port Alice.

The number of fry released in 1998 (399,700) was low as a result of poor brood capture conditions in 1997. High water levels in the fall and low natural escapement numbers reduced the regular capture numbers (Appendix XI). Brood stock collection conditions were much better in 1998 due to lower flows and closures or restrictions on salt water fishing which increased escapement numbers.

The Forestry Department costs associated with the Salmonid Enhancement Program were \$8,442. An additional \$70,695 came from the Company's Public Projects fund. The federal Department of Fisheries and Oceans reimbursed \$15,000 of the Public Projects expenditures.



Shelagh Fox and Jim Basha assist at the Cordy Creek Hatchery with the 1998 coho brood stock egg takes. Coho stock returns in TFL 6 rivers were very strong due to restricted commercial and sports fishing. As a result, the Cordy Creek Hatchery was successful in capturing enough broodstock for 440,000 eggs.

3.2.7.2.1 Cordy Creek – Holberg

Chinook, coho, and chum salmon fry were released into the Goodspeed River, and coho salmon fry were released into the San Josef River. A total of 90,397 fry were released from the Cordy Creek hatchery (Appendix XVI).

The Environmental Youth Team work crew sponsored by Western Forest Products and the Ministry of Environment provided labour for maintenance work at the Hatchery. The crew painted the hatchery and tower, constructed cap trough covers, and cleared trails to facilitate brood stock collection.

3.2.7.2.2 Marble River – Port McNeill

The Friends of the Marble River manage the Marble River Hatchery, south of Port McNeill. Poor brood stock capture numbers in 1997 led to low numbers of fry for release: 201,400 chinook, 37,200 coho, and 3,200 steelhead fry were released into the Marble River watershed (Appendix XVI).

Work started on the rearing channel project. The outflow channel and pond were excavated in the spring, the intake box installed during the summer fish “window” and the pipeline connection to the channel in the fall. Funding for the project was received from the Department of Fisheries and Oceans, the Quatsino Sound Salmon Enhancement Society and the Pacific Salmon Foundation. In-kind support was received from Western Forest Products and many local contractors.



Grant Anderson and his wife Debbie have been heavily involved in managing the Marble River Hatchery for the past seven years. In the spring of 1998 a total of 201,400 chinook fry were released using helicopters and truck mounted tanks to restock streams in the Marble River watershed.

3.2.7.2.3 Colonial Creek – Jeune Landing

The Port Alice Fish and Wildlife Association, that includes many Western Pulp and Western Forest Products employees, continued to operate the Colonial Creek Hatchery. The Colonial Creek

Hatchery raised 67,520 coho and chinook salmon for release in the spring of 1998 (Appendix XVI). The Environmental Youth Team crew, sponsored by Western Forest Products, assisted in repair and maintenance of the hatchery facilities.

3.2.8 Wildlife Habitat

Special measures were taken in one Jeune Landing cut block to address wildlife values (Table 8). Western Forest Products' management approach is shifting from individual species to a landscape level approach where defined habitat types will be maintained to provide for a range of sensitive species. The Company will continue to inventory and individually manage special wildlife features including raptor nests and bear dens.

3.3 Integration of Harvesting Activities

Western Forest Products encourages the use of TFL 6 by other stakeholders. Mushroom and salal pickers have access to our areas and are only restricted by safety and fire regulations. Yew bark harvest is encouraged in areas under cutting permit. Agreements have been reached with First Nations to focus harvest of traditional materials, mainly cedar bark, in cutting permit areas. Beekeepers use our roads and recreation use is encouraged through the maintenance of existing recreation sites and the expansion of others. However, commercial use of existing public sites is not permitted. Trapline holders are notified when new cutting permits are issued.

3.4 Forest Fire Management

3.4.1 Prevention

Fire Preparedness Plans were submitted to the Ministry of Forests in the spring of the year by all Western Forest Products Operations. Holberg Operation maintained four weather stations, Jeune Landing Operation, two weather stations; and Port McNeill Operation, three weather stations through the fire season. Total forestry costs for fire prevention and suppression were \$12,351.

3.4.2 Suppression

Jeune Landing and Port McNeill reported no accidental fires requiring suppression in the year. Holberg Operation suppressed three accidental fires, which affected 5.9 ha of land (Appendix XI).

3.4.3 Fuel Management

The last TFL 6 fuel management plan was produced in 1995 and covered the period from 1996 through the year 2000. This plan uses fuel type, climatic zone, aspect, and slope to establish a hazard rating that is used to determine fuel management practices. Slash and pile burning are generally not required for hazard abatement purposes. Roadside burning was carried out in 1998 for silvicultural purposes.

3.5 Forest Health Management

Forest health is monitored as part of other regularly scheduled silviculture surveys. Specific forest health costs are only assigned when there is a specific forest health situation such as an insect outbreak.

3.5.1 Disease Management

No unusual disease issues were identified during the year. Hemlock dwarf mistletoe and various root rots were noted during silviculture surveys or when preparing silviculture prescriptions and stand management prescriptions. Formal surveys and special management techniques are employed when unusual situations are encountered. None were identified in 1998.

3.5.2 Insect Management

The western black-headed budworm (*Acleris gloverana*) outbreak continued to develop within the TFL. Aerial surveys conducted with the Ministry of Forests showed the majority of the infestation in the Holberg Operation to be light. Most of the Jeune Landing infestation was in vigorous second growth stands thus little long term impact is expected. There was little development in Coal Harbour or around Port McNeill. Costs to monitor the outbreak were \$3,863.

Ambrosia beetle (*Gnathotrichus* spp. and *Trypodendron lineatum*) trapping continued at the Port McNeill Dryland Sort. Captures of both species were significantly reduced from 1997 levels (Table 11). Windfall in the vicinity of the Port McNeill Dryland Sort may have attracted many of the emerging adults in the spring. The costs of the ambrosia beetle trapping program are absorbed into logging costs.

Table 11: Ambrosia Beetle Trapping Summary – Port McNeill Dryland Sort

Species	Year		
	1996	1997	1998
<i>Gnathotrichus</i> spp.	1534	2293	750
<i>Trypodendron lineatum</i>	463249	982500	348525
TOTAL	464783	984793	349275

3.5.3 Ungulate Management

Deer browsing of regeneration is not a significant problem in most areas of TFL 6. No browse guards or repellents were installed or maintained in the year. During the year the District Manager

officially designated eleven of Western Forest Products' 14 Deer Winter Ranges as Ungulate Winter Ranges.

3.5.4 Abiotic Factors

Several areas of windthrow were detected in the Holberg Operation during the year. Silviculture Plans were submitted for 7 salvage areas. One harvested block at Holberg Operation was modified to reduce the effect of windthrow. Port McNeill Forest Operation continued to use a helicopter pruning bar to reduce the sail on individual trees in windthrow prone reserve areas.

3.6 Silviculture

3.6.1 Reforestation

3.6.1.1 Seed Acquisition

Lost Lake Seed Orchard in Victoria and the Saanich Forestry Centre in Saanichton are owned and operated by Western Forest Products. They are licensed by the Ministry of Forests to produce genetically superior seed and stockings for reforestation on managed forest lands. Seed orchards managed for seed production include two western red cedar orchards, two western hemlock orchards, one Sitka spruce orchard, one coastal Douglas-fir orchard, one yellow cypress clonal orchard, and one cottonwood clonal orchard. No wild stock seed collections were made in TFL 6 during the year.

Approximately 20.9 hectolitres of cones were harvested from the western red cedar orchards with an estimated potential of 2.8 million seedlings. The western hemlock seed orchard yielded 2.3 hl of cones with an estimated potential of 0.4 million seedlings. Fifteen percent of the western hemlock seed came from elite crosses.

All of the coastal Douglas-fir seed collections resulted from elite crosses. The 1.5 hl of cones harvested should provide enough seed to produce 28,000 seedlings. This high gain custom seedlot is projected to yield a 17.5 per cent volume gain at rotation. Costs for this project were \$7,854 and were covered by Forest Renewal BC through the Operational Tree Improvement Program. The seed production summary is included in Appendix XXII.

Cuttings from the yellow cypress clonal hedges yielded 300,000 stockings. The cottonwood stool beds produced 2,800 whips. The TFL 6 prorated share of seed collection and acquisition costs were \$151,498. The prorated share of seed sales was \$22,973.

Dr. Ruth Nusbaum, (SGS Forestry, Oxford, England) Dr. Yousry ElKassaby, (former Director of Western Forest Products' Saanich Forestry Centre) and Len Apedale (SGS Forestry, Canada) tour the Company's seed orchard operations as part of the pre-assessment for the Company's Forest Stewardship Council certification.



3.6.1.2 Tree Improvement

Orchard management included consolidation of the second-generation western hemlock orchards. High breeding value clones were removed from Lost Lake Seed Orchard (Orchard Hw 176) and incorporated into the Saanich Forestry Centre Seed Orchard (Orchard Hw 170). The projected volume gain at rotation was thereby increased from 7 per cent to 14.5 per cent. Costs for this project were \$19,146. Funding was received from Forest Renewal BC through the Operational Tree Improvement Program. A number of orchards were removed from the Saanich Forestry Centre, including one and a half coastal Douglas-fir orchards, a western white pine orchard, and a yellow cypress orchard.

Special management techniques were applied to enhance genetic gain at the seed orchards. Polycross breeding of selected western red cedar clones produced seed for progeny testing. Forest Renewal BC through the Science Council of BC (\$24,000) funded this work.

Phase 1 of the Second Generation Western Red Cedar Seed Orchard is underway. To date, 150 trial clones were re-propagated by rooting of cuttings. These will be rogued after results of the progeny trials are analysed. Those clones with high breeding values will be included in the second-generation orchard. Forest Renewal BC funded the program for \$15,400.

Gibberellic acid was applied to approximately 600 trees in two western red cedar orchards to induce seed production. This will provide several years' supply of class A seed. Forest Renewal BC funded this program (\$13,970) through the Operational Tree Improvement Program.

Field testing of the clonal yellow cypress hedges continued in 1998 with the establishment of two trials at Port McNeill. Four seedlots represented by 250 clones were included in the trials. Data

collection and analyses of earlier trials continued. The final phase of testing will be established in the spring of 1999. Holding areas for the trial material was established at Saanich Forestry Centre. Costs for this project, funded by Forest Renewal BC through the Science Council of BC, totalled \$80,000.

Evaluation of the yellow cypress clonal material has led to the establishment of a High Gain Yellow Cedar Clonal Orchard. To date the top 46 clones out of 1,700 tested were selected for inclusion and are now being rejuvenated in a nursery. The average computed volume gain at rotation, based on 7-year data, is 18 per cent. Costs for this project, funded by Forest Renewal BC through the Operational Tree Improvement Program, were \$13,440.

3.6.1.3 Site Preparation

Site preparation tools on north Vancouver Island include broadcast burning, pile burning, and mechanical site preparation. These techniques are employed to improve plantability or reduce brush competition. Appendix XI details the 1998 site preparation program. Costs for site preparation in TFL 6 were \$123,761.

3.6.1.3.1 Broadcast Burning

There was no broadcast burning in TFL in 1998. Smaller cutblock sizes and the increased number of riparian reserves have made prescribed burning both difficult and expensive. This is only the second year in the past thirty where broadcast burning was not used as a silvicultural tool.

3.6.1.3.2 Pile Burning

Roadside debris was piled and burned at all Operations in TFL 6: Port McNeill completed 24.1 km of roadside burning, Holberg completed 58.6 km, and Jeune Landing completed 6.5 km for a total of 89.2 km (Appendix XI). Average cost was \$561 per km.



Pile burning was completed on K500 landings in cut block 804 in October. Pile burning is carried out to increase the number of plantable spots for reforestation adjacent to roads.

3.6.1.3.3 Mechanical Site Preparation

Holberg and Port McNeill used excavators to prepare planting spots on 25.6 ha of harvested area (Appendix XI). Costs for mechanical site preparation averaged \$1,236 per ha.

3.6.1.4 Planting

Silviculture contractors planted 1.47 million seedlings on TFL 6 during the year. Original plantations were established on 1 333.7 ha and fill-plants or replants were carried out on an additional 55.3 ha of previously regenerated area (Appendix XI). The NSR area recorded on the TFL 6 Regeneration Balance sheet decreased significantly during the year (Appendix XII). Appendix XIII summarizes the planting history of TFL 6.

Fertilization at time of planting improves the competitiveness of conifer seedlings on salal-dominated sites. Western Forest Products Limited fertilized 485,000 seedlings at time of planting in 1998.

The bulk of the seedlings planted on TFL 6 were western red cedar (37 per cent). Other major species were amabilis fir (28 per cent) and western hemlock (23 per cent). The planting program also included small numbers of cypress, Douglas-fir, Sitka spruce and shore pine.

Costs for the planting program totalled \$1,438,352. This averages to \$1,025 per ha or \$0.97 per tree. This included stock purchase and inspection, planting costs and fertilization at time of planting costs. Forest Renewal BC's Backlog Program reimbursed the Company \$17,427 for areas logged prior to October 1, 1987 that remain a government responsibility.

3.6.1.5 Stocking Surveys

Surveys of naturally regenerated areas at Holberg, Jeune Landing, and Port McNeill showed high success rates. Only 7.4 ha of 190.0 ha surveyed, less than 4 per cent of the area, was not satisfactorily restocked (Appendix XI). A small area at Jeune Landing was classified as non-productive forest land. The natural regenerated area is credited to the regeneration balance sheet (Appendix XII).

Ninety-eight percent of the plantation area surveyed met or exceeded TFL 6 stocking standards (Appendix XI). Only 42.8 ha of a surveyed 2 358.3 ha was not satisfactorily regenerated. The NSR area is debited to the regeneration balance sheet (Appendix XII) and scheduled for treatment. The Not Satisfactorily Regenerated Balance for TFL 6 was 1 601 ha as of December 31.

Costs for stocking surveys in TFL 6 were \$67,925 or \$27 per ha. The Forest Renewal BC reimbursements for areas logged before October 31, 1987 were \$1,632.



Mr Herb Doman, Chairman and President of Western Forest Products, assisted by Port McNeill Junior Forest Wardens, completes the planting of the Company's 75 millionth seedling at Rupert Arm Recreation Site.

3.6.1.6 Planting Survival Assessments

Second year survival assessments of planted areas showed high survival rates, exceeding 91 per cent for all Operations. Survival assessments were conducted on a total of 1 444.1 ha of TFL 6 (Appendix XI).

Survival assessments cost \$41,790 or \$29 per ha. Forest Renewal BC reimbursements for areas logged before October 31, 1987 were \$2,398.

3.6.1.7 Free Growing Surveys

All Operations carried out free growing surveys during the year. A total of 11 304.5 ha were surveyed. Fifty-eight percent of the surveyed area (6 508 ha) met Management Plan free growing standards. Only minor treatments are necessary to declare the balance free growing. Surveyors classified 23.9 ha as "other" (Appendix XI).

Costs for free growing surveys were \$77,807 or \$7 per ha. Forest Renewal BC reimbursed the Company \$72,665 for areas harvested prior to October 31, 1987.

3.6.2 Stand Management

3.6.2.1 Brushing and Weeding

Silviculture contractors carried out brushing and weeding treatments on 1 074.0 ha of forest land. A variety of brushing and weeding techniques successfully reduced alder and salmon berry

competition (Appendix XI). Costs of the brushing and weeding program were \$804,140. Forest Renewal BC reimbursed \$696,498 for areas harvested prior to October 31, 1987. Costs rose dramatically due to the uncertainty introduced by New Forest Opportunities involvement in the brushing program.

3.6.2.2 Juvenile Spacing

All Operations implemented small spacing programs during the year (Appendix XI). Silviculture contractors spaced a total of 76.4 ha of dense regeneration on TFL 6. Costs for spacing were \$183,574 or \$2,403 per ha. Forest Renewal BC funded the projects through the Enhanced Forestry Program. The reduced spacing program resulted from bureaucratic delays in the implementation of the Forest Renewal BC Multi-Year Agreement. Costs rose dramatically due to New Forest Opportunities' involvement in the program.

3.6.2.3 Pruning

All Operations implemented small pruning programs during the year. Silviculture contractors pruned 256.6 ha of previously spaced stands (Appendix XI). The reduced program in 1998 reflected the bureaucratic delays in implementing the Forest Renewal BC Multi-Year Agreement. Costs for pruning totalled \$607,947, or \$2,369 per ha. Costs rose over historic levels due to New Forest Opportunities' involvement in the program.



Juvenile spacing continues to be an important silvicultural practice in young forest stands in TFL 6. This hemlock stand off Port Alice Main on Port Alice 10, established in 1985, was spaced in 1997 to 725 stems per ha. Of those, 408 stems per ha were pruned.

3.6.2.4 Fertilization

Forest Renewal BC cancelled funding for the large fertilization program that was prepared for spring 1998. As a result there were no broadcast applications during the year. Total fertilization expenses were \$24,910 to prepare for an extensive program in 1999. Forest Renewal BC reimbursed the Company \$18,747 under the Enhanced Forestry Program.

3.7 Roads and Bridges

3.7.1 Construction

Road building in 1998 included 66.5 km of new construction and 17.3 km of rebuilt road (Table 12). Fourteen bridges were constructed including a 27.5 m steel and concrete crossing of the Utluh in Jeune Landing, and a 46 m steel and concrete structure on Quatsino Main at Holberg. Road construction was halted for the year on July 10th in response to deteriorating markets.



Construction of the Quatsino Main bridge opens access to 1.5 million cubic metres of timber. Sections were pulled across rollers on the temporary support frame to put the bridge in place.

Table 12: Road and Bridge Construction
Summary

Operation	New Roads (km)	Rebuilt Roads (km)	Bridges (#)
Holberg	31.2	13.9	5
Jeune Landing	20.9	2.8	1
Port McNeill	14.4	0.6	8
TOTAL	66.5	17.3	14

3.7.2 Maintenance

Seven hundred and sixty-eight kilometres of road were maintained in TFL 6. Appendix XI provides a summary by Operation.

3.7.3 Current Deactivation

Seasonal, semi-permanent, and permanent deactivations of roads are obligations in current areas. Operations deactivated 53.4 km of road during the year: 20.3 km was seasonally deactivated, 19.5 km was semi-permanently deactivated, and 13.6 km was permanently deactivated. Appendix XI lists the amount of deactivated road by level of deactivation and Operation.

3.8 Employment and Economic Opportunities

3.8.1 Direct Employment

The following sections outline direct forest management and manufacturing employment for TFL 6. The manufacturing employment is for the TFL 6 volume processed in Western Pulp or Doman mills and does not include direct employment generated through log sales or trades.

3.8.1.1 Planning, Engineering, and Road Development

Planning for TFL 6 took place at each Forest Operation, in the Port McNeill Regional office and at the Company's head office in Vancouver. Planning and development at the regional level included implementation of the watershed restoration program and local contract mapping for development plans. Planning employed 25 company and contract employees for 1,300 days (Appendix XV – A).

Engineering and road construction employed 106 people and provided an estimated 12,409 person days of employment at Port McNeill, Jeune Landing, and Holberg Forest Operations and the Port McNeill Regional office. Twenty per cent of the employment was Company personnel, with the remainder contractors. Employment was down significantly in 1998 due to suspension of road construction on July 10th in response to poor market conditions.

3.8.1.2 Harvesting

Harvesting employment includes the support staff at each Operation. Ninety-five percent of the 487 people employed in harvesting are local residents and 56 percent are contract employees. TFL 6 harvesting person-days totalled 60,160 for the year (Appendix XV – A). This is down significantly from 1997 due to production curtailments forced by poor markets. With minor variations, Forest Operations operated from January 26th to July 10th and October 4th to November 12th. The Quatsino Dryland sort operated from January 22 to August 12th and October 26th to December 11th.

3.8.1.3 Transportation

Logs were barged or towed from northern Vancouver Island to processing facilities and markets in the south. Log transport generated 7,071 person-days of employment (Appendix XV – A).

3.8.1.4 Processing

Eight Doman and Western Pulp mills processed wood from TFL 6. These were Duke Point Sawmill, Ladysmith Sawmill, Cowichan Bay Sawmill, Silvertree Sawmill, Vancouver Sawmill, Nanaimo Sawmill, Nanaimo Log Merchandising, and the Port Alice Pulp Mill. Chips from the sawmills were sent to WP's Squamish Pulp Mill. A portion of the TFL 6 volume was either sold or traded to other forest companies or manufacturers.

Prorated direct employment based on wood flow and consumption estimates totalled 75,628 person-days (Appendix XV – A). This total does not include employment generated by TFL logs that were sold or traded to other manufacturers.



One of four 'new' digesters for installation at the Port Alice Pulp Mill acquired from Ketchikan Pulp Co. after it closed in March 1997. The \$25 million project included moving and installing four digesters, evaporators, and baling equipment. This equipment reduced energy and chemical costs, improved yield and pulp quality, and brought the mill back to 470 tonnes per day operating rates.

3.8.1.5 Silviculture and Integrated Resource Management

Basic and enhanced silviculture, and integrated resource management projects employed 334 people throughout the year at the Operations and Region. Ninety-two percent were employed by contractors. More than 40 percent of the basic and enhanced silviculture employees were local residents who generated more than 60 per cent of the person-days (Appendix XV – A). The TFL 6 prorated share of the Saanich Forestry Centre employment (46 full time and seasonal employees) was 1927 person-days. Silviculture and Integrated Resource Management generated 13,377 person-days of employment in the year.

Administrative delays in implementation of the Forest Renewal BC Multi-Year Agreement and the added costs and uncertainty associated with New Forest Opportunities Limited reduced silviculture employment. Western Forest Products' policies continued to focus on the development of First Nation and other local skilled crews.



Western Forest Products' employees at the Saanich Forestry Centre package 1-year-old cedar seedlings for reforestation in TFL 6. The Company now has a new 4 million tree nursery at the Saanich Forestry Centre as part of the acquisition of Pacific Forest Products in late 1997.

3.8.1.6 Administration

Administration of the TFL is carried out from the Forest Operation, Regional and corporate offices. Forty Company employees spent a portion of their time in the administration of TFL 6. The prorated share of administrative employment for TFL 6 was 4,232 person-days for the year.

3.8.2 Indirect Employment

TFL 6 activities support secondary jobs in the local communities of Holberg, Port Alice, Port McNeill, Coal Harbour, Quatsino, Winter Harbour, and Port Hardy. While these communities are slowly diversifying their economies, they all still rely heavily on the forest industry.

Vancouver Island and lower Mainland communities also benefit from the timber produced from TFL 6 through secondary jobs. Significant income is generated in the processing locations: Nanaimo, Ladysmith, Cowichan, and Vancouver. Additional indirect and induced jobs are supported province-wide through services and supplies.



Don McGee, the first professional forester in TFL 6 in the early 1950s, and Bill Dumont, Western Forest Products' Chief Forester, attended a community reunion in Port Alice during the year.

Indirect job opportunities are supported by TFL 6 through the spending of wages from direct jobs. This spending supports jobs in the service, retail, finance, and housing industries.

The Provincial Government stumpage and royalty revenue from this tenure was about \$26 million in 1998. Both the Federal and Provincial Governments collected significant revenue from income and other taxes as a result of TFL 6 logging, forest management, and timber processing. Municipal Governments also receive revenue through taxes from TFL 6 operations within their jurisdictions. This revenue supports indirect jobs within Government Ministries and agencies at all levels.

The ratio of direct employment to indirect and induced employment used in this analysis is 1:2 (Price-Waterhouse). For every employee directly supported by TFL 6 activities, two additional people are indirectly employed through private and public sector agencies. The estimated number of person-days of indirect and induced employment generated by TFL 6 activities in 1998 was 348,354. At 180 person-days per year, TFL 6 activities indirectly supported more than 1,900 full-time equivalent jobs. This is a conservative estimate, as the direct employment generated by TFL 6 was reduced by abnormal shutdowns at Mills and forest operations during the year.

3.8.3 First Nations Employment and Initiatives

The Company continued to work closely with the Quatsino First Nation on their phase logging contract. A total of 35 000 m³ were harvested by the Quatsino Forestry Company. The Quatsino First Nation has developed a productive, safe, and efficient operation. Western Forest Products continued to provide a First Nations' silviculture contractor endorsed by the Quatsino Band Council with direct awards for brushing, pruning, and juvenile spacing. In response to a request from the Quatsino First Nation, Western Forest Products agreed to assist with the management of Woodlot Number 72 which is adjacent to TFL 6.

The Company continued to work on an expanded relationship with the Kwakiutl First Nation based in Fort Rupert. Western Forest Products supported training initiatives and provided a series of direct awarded silviculture contracts for brushing, revegetation, and slash burning to a silviculture contractor endorsed by the Kwakiutl Band Council.

Silviculture contracts in TFL 6 provided 1987 person-days of First Nation employment in 1998. This was more than 31 per cent of the contract silviculture work on the tenure. The Company target of providing a minimum of 20 per cent of the silvicultural contract employment on a tenure to First Nations was exceeded. Appendix XV -C provides First Nations silviculture contract information.

3.9 Performance Monitoring

The Ministry of Forests carried out 174 harvest inspections in TFL 6 during the year. All were found to be in compliance and only minor instructions were issued. Four Pre-Determination Hearings involving alleged non-compliance incidents that occurred in 1997 were held. By year-end, the Determinations were still outstanding.

The Company also undertakes internal audits using an external team of experts in engineering and forestry. A total of 9 cutblocks and 16.7 km of road were reviewed at the Port McNeill and Coal Harbour Forest Operations. The audits found roads to be built to high standards with road width, culvert size, and culvert placement matching prescriptions. Cut blocks were considered well managed after logging except for minor concerns with follow-up maintenance of ditches and culverts. It was recommended that ditch water be directed away from areas with inherently poor drainage. Other issues such as windthrow in riparian areas were also noted. Action Plans were prepared for the audits and outstanding items addressed.

Forestry Department costs in support of audits totalled \$4,382.

4.0 TIMBER PROCESSING

An estimated 757 000 m³ of TFL 6 logs were processed at seven Company owned mills on the lower mainland and southern Vancouver Island as well as at the Port Alice Pulp Mill on Northern Vancouver Island. An additional 111 000 m³ of TFL 6 logs were sold or traded on the Vancouver Log Market. Table 13 summarizes log flow for TFL 6.

Table 13: TFL 6 Log Consumption

Processing Facility	Log Volume (m3)	Portion of 1998 Consumption (per cent)
Port Alice Pulp Mill	176 000	31
Duke Point Sawmill	259 000	95
Ladysmith Sawmill	111 000	28
Silvertree Sawmill	80 000	23
Vancouver Sawmill	44 000	13
Nanaimo Log Merchandizer	31 000	9
Nanaimo Sawmill	33 000	7
Cowichan Sawmill	23 000	6

The Chemainus Value-Added Mill re-manufactured 130 thousand cubic metres of wood during the year. Some of this volume came from these facilities. The Squamish Pulp Mill consumed 608 thousand units of chips. An estimated twelve per cent of this volume came from TFL 6 logs manufactured at these mills.

Estimates of log flow and wood consumption are summarized by tenure in Appendix XVII.



Hi-grade Doman lumber at the Company's Value-Added plant in Chemainus. Lumber inventories exceeded 500 million board feet by mid-year, resulting in extended down time for logging operations until the lumber was sold.

5.0 RESEARCH

Western Forest Products supported numerous research projects in TFL 6 throughout the year. Data collection and reports completed in 1998 are highlighted in the following sections. All active projects in the TFL, including those not scheduled for monitoring in the year, are listed in Appendix XVIII.

Forestry Department research costs totaled \$297,789 including the pro-rated share of the Saanich Forestry Centre's genetics research program. Reimbursements included Ministry of Forests Direct Contracts, and Forest Renewal BC funds.

5.1 Silviculture and Stand Management Research

5.1.1 Salal Cedar Hemlock Integrated Research Program (SCHIRP)

Salal-dominated sites on northern Vancouver Island have been the subject of the SCHIRP program for more than a decade. Numerous organizations have cooperated on related studies. Findings from this Program have led to an extensive operational fertilization program on the TFL.

All post-1997 growing season measurements were completed in early 1998 before spring flush. This included operational fertilization monitoring plots and the 1996 SCHIRP Installations. Salal measurements were repeated in the 1996 SCHIRP Installations. Following the 1998 growing season, the demonstration trial measurements and the 1997 Operational Fertilization Monitoring Plot measurements were repeated.

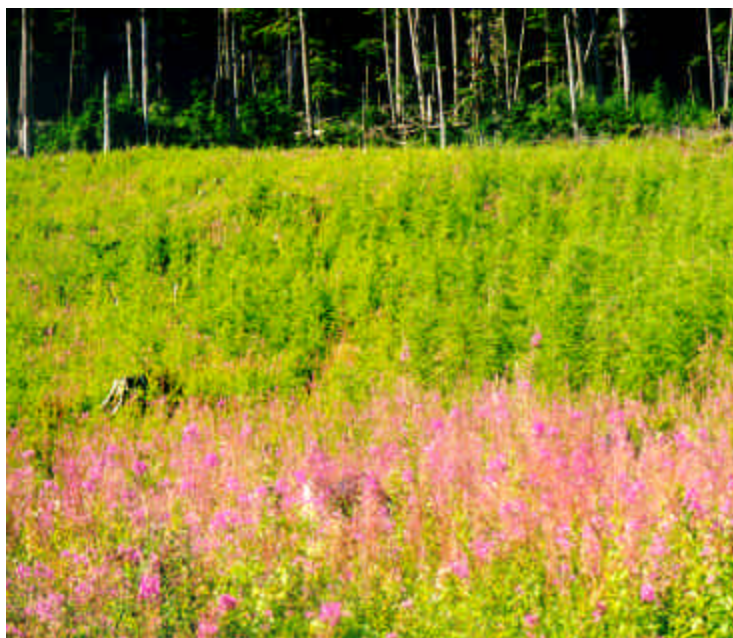
The poor performance of conifers on mid-elevation sites dominated by blueberry and huckleberry was examined during the year. Western Forest Products will be cooperating with the Canadian Forest Service in establishing vegetation management trials on these Vaccinium sites. One of two sites in this trial will be at Holberg within TFL 6.

5.1.2 Drainage Trial

Maintenance and monitoring of the Suquash Main drainage trial continued throughout the year. Seedlings were planted in the areas disturbed by the trial areas and were measured and added to the monitoring plots. All seedling heights were measured after the 1998 growing season. Water level monitoring continued throughout the year.

5.2 Genetics Research

The yellow cypress clonal evaluation trials at Port McNeill, Holberg, and Jeune Landing continued to be monitored and maintained throughout the year. New installations were established on Misty Main and Rupert 600 for the eighth phase of the evaluations. Plans were in place and sites selected for the ninth and final phase of the evaluations.



On Pemberton 400 at Coal Harbour, fireweed provides an attractive foreground to the Hemlock Realized Gain Trial. This area was logged in 1991 and planted in 1992 with 5,110 genetically improved trees.

The second phase of the yellow cypress trials (Coal Harbour) was measured after the 1998 growing season. This was the seven-year evaluation. The sixth phase (Port McNeill), after four growing seasons, was also measured.

5.3 Forest Inventory

The Site Index Conversion Survey continued from 1997. This project, contracted to Western Forest Products by the Ministry of Forests, targeted young mixed stands: western hemlock and amabilis fir, western hemlock and western red cedar, and amabilis fir and western red cedar.

Western Forest Products staff prepared a proposal to begin the re-inventory of TFL 6 under the Forest Renewal BC under Operational Inventory Program. This Vegetation Resource Inventory will update the current forest information with a focus on the contribution made by developing second growth stands. The field portion of the VRI will start in 1999.

6.0 GOALS AND INITIATIVES

Western Forest Products manages TFL 6 consistent with the objectives outlined in Management Plan 8. Appendix XX outlines progress made in addressing Management Plan 8 commitments and issues in 1998. For 1999 there are specific silviculture, planning, inventory, research, recreation, First Nations, and market goals.

The silviculture goals are to complete all basic silviculture obligations according to operational plan schedules as well as to pursue opportunities to expand the current enhanced silviculture programs. Program details are provided in Table 14.

Table 14: Goals for 1999

Program	Project	Target
Silviculture	Site Preparation	182 ha
	Planting	1 448 ha
	Survival Surveys	1 740 ha
	Stocking Surveys	2 440 ha
	Brushing	862 ha
	Free Growing	4 200 ha
	Spacing	118 ha
	Pruning	133 ha
	Fertilization	1 692 ha
Planning	De-centralize FDP process	TFL 6 Blocks 1 and 2
	De-centralize mapping services	TFL 6 Blocks 1 and 2
	Landscape Unit Analysis	TFL 6 Blocks 1 and 2
	Start MP 9	TFL 6 Blocks 1 and 2
Inventory	Wildlife Modelling	Special Management Zones
	Fisheries Inventory	Special Management Zones
	VRI Phase 1	TFL 6 Blocks 1 and 2
Research	SCHIRP Monitoring	TFL 6 Blocks 1 and 2
	Suquash Monitoring	TFL 6 Block 2
	CRIC (phase 9) Installation	TFL 6 Block 2
	Windthrow Research	TFL 6 Blocks 1 and 2
Recreation	SMZ # 2 Development	TFL 6 Block 1
	Recreation Site Maintenance	TFL 6 Block 1 and 2
First Nations	Capacity Building	TFL 6 Blocks 1 and 2
Certification	FSC, ISO, CSA	TFL 6 Blocks 1 and 2

Planning goals include administrative efficiencies of landscape level programs. Plans are underway to complete the decentralization of the Forest Development Planning process to individual Forest Operations and move FDP mapping from the Company's corporate office to local contractor(s). Preparation of the data package required to support Management Plan 9 has begun. This data

package and analysis will support future implementation of the Landscape Unit objectives of the Vancouver Island Land Use Plan.

Forest inventory efforts will be concentrated on updating second growth information while other inventory efforts will be focussed on fish, wildlife, and other biodiversity values within Special Management Zones. In 1999, phase 1 of the re-inventory of TFL 6 forests and the wildlife and fisheries projects started in 1998 will be completed.

Current research projects will be maintained. There will be continued monitoring of SCHIRP sites; installation of mid-elevation SCHIRP style installations to examine Vaccinium competition, continued monitoring of the Suquash drainage site, and further research into windthrow management and estimates of windfall losses.

Recreation goals will be to maintain existing Western Forest Products' sites within TFL 6 as well as to continue cautious development of Special Management Zone # 2 while keeping in mind the Special Management Zone recreation objectives. There will be an attempt to resolve Raft Cove public access issues.

First Nations efforts will be concentrated on continued capacity building with both the Quatsino and Kwakiutl First Nations. The Quatsino First Nation currently logs, implements silviculture treatments, and completes assessments for Western Forest Products. The Kwakiutl First Nation currently implements silviculture treatments for the Company.

Certification will remain a high priority with Western Forest Products for 1999. The Company will continue its efforts to obtain FSC, ISO and CSA certification for TFL 6 woodland operations in order to aid marketing of forest products in foreign markets.

7.0 ADMINISTRATION

The costs of supervision and overhead relating to specific operational projects are included in the reported project costs. However, many administrative costs are broadly defined as general management and overhead. These include Company prorated charges to the Forestry Department for rentals and services as well as salaries and expenses to perform administrative functions. Total costs for administration were \$682,439.

8.0 FINANCIAL STATEMENTS

8.1 Forest Management Costs

The following summary presents all TFL 6 Forest Management Costs incurred by Western Forest Products and other agencies. Planning, supervision, and overhead costs directly related to programs as well as program application costs are included before credits or reimbursements. Head office forestry department costs have been included as either a direct charge for specific services or as a prorate based on Company volume for administration and planning.

Program	Section		Sub-Total
Planning			647,913
	Higher Level Planning	2.4.1	60,850
	Forest Development Planning	2.4.2	143,966
	Cutting Permits and CP Cruising	2.4.3.1	274,172
	Silviculture Prescriptions	2.4.3.3	168,925
Public Involvement			96,760
	Plan Viewings	2.5.2	6,433
	Forest Education	2.5.5	90,327
Inventories and Mapping			472,160
	Geographic Information System	2.6.1	84,188
	Forest Inventory	2.6.2	168,559
	Ecosystems	2.6.3	14,911
	Terrain Stability	2.6.4	5,820
	Resource Inventories & Assessments	2.6.6	198,682
Utilization			123,486
	Residue Assessments	3.1.3.2	75,817
	Minor Products	3.1.3.3	43,169
	Commercial Thinning		4,500
Conservation and Protection			245,633
	Operational Site Stabilization	3.2.3.2	70,615
	Recreation Resources	3.2.5	75,285
	Salmon Enhancement Program	3.2.7.2	79,137
	Fire Management	3.4	12,351
	Forest Health	3.5	3,863
	Audits	3.9	4,382
Silviculture			3,521,704
	Seed Procurement	3.6.1.1	151,498
	Site Preparation	3.6.1.2	123,761
	Planting	3.6.1.3	1,438,352
	Stocking Surveys	3.6.1.4	67,925
	Survival Assessments	3.6.1.5	41,790
	Free Growing Surveys	3.6.1.6	77,807
	Juvenile Spacing	3.6.2.1	183,574
	Brushing & Weeding	3.6.2.2	804,140
	Pruning	3.6.2.3	607,947
	Fertilization	3.6.2.4	24,910
Research			297,789
	Research	5.0	297,789
Administration			639,152
	Administration	7.0	682,439
TOTAL			\$5,188,145

8.2 Forest Management Reimbursements

Western Forest Products received funding from government agencies for silvicultural enhancement, integrated resource management, and salmonid enhancement projects. A summary of the reimbursements received is presented below.

Program	Project	TFL 6	TFL 19	TFL 25	TOTAL
Enhanced Forestry	Juvenile Spacing	141,896	356,177	501,306	999,379
	Pruning	553,606	148,701	689,483	1,391,790
	Fertilization	12,855	10,029	14,869	37,753
Forest Renewal BC					
Backlog Forestry	Surveys	72,871	31,679	4,150	108,700
	Planting	17,427	3,046		20,473
	Brushing	696,062	6,685	313,474	1,016,221
Forest Renewal BC	Site Rehabilitation		34,017		34,017
Recreation					
Recreation	Recreation	24,942	2,208	7,780	34,930
Forest Renewal BC					
Operational Inventory	Cultural Heritage		8,551		8,551
	Streams	68,893	16,514		85,407
	Timber			116,320	116,320
Forest Renewal BC	Wildlife	38,427	78,326	105,482	222,235
	Water Quality		34,017		34,017
Science Council	Research	330,268	121,129	82,878	534,275
	Seed Orchard Development	22,973	11,486	7,859	42,318
Forest Renewal BC					
Department of Fisheries and Oceans	Salmonid Enhancement Program	15,000			15,000
Ministry of Forests	Brushing		7,288		7,288
	Planting		23,286		23,286
	Surveys			9,316	9,316
	Recreation		94,979	9,747	104,726
	Research	13,635	2,280	1,560	17,475
South Moresby FRA	Juvenile Spacing			247,086	247,086
	Wildlife Inventory			75,000	75,000
TOTAL		2,008,855	990,398	2,186,310	\$5,185,563

APPENDICES

Appendix I

TREE FARM LICENCE 6
1998 Scaled Production

CUBIC METRES

Operation	Mark	Volume	Total
Botel Lake	6/300	243.4	
	6/301	25248.4	
	6/302	22284	
	6/5	0	
	6/800	97.5	
	6/801	8613.3	
	6/803	17102	
	6/870	-21.1	
	6/871	-3.3	
	6/873	5582.8	
	NAXNQ	1402.5	
	T0660A	14170.9	94720.4
Coal Harbour	6/600	5.6	
	6/601	41706.2	
	6/602	0.8	
	6/61	334.9	
	JZ004	43.7	
	KZ003	108	
	T0457A	0.8	
	T0475A	14.2	
	T0475B	0	
	T0693A	-1677.8	
	T0693B	2156.9	
	T0700A	114.1	42807.4
Holberg	6/100	25.1	
	6/102	7211.8	
	6/104	108587.5	
	6/750	38739.6	
	6/800	34.4	
	6/801	255.3	
	6/802	12885.8	
	6/803	11139.4	
	6/804	35502.5	
	6/870	0.2	
	6/871	-120.3	
	6/872	408.5	
	6/873	24.2	
	6/90	5113.8	
	CZ010	9281.5	
	LL44176		
	T0386A	4273.2	
	T0386B	5720.6	
	T0391A	2554.9	
	T0399A	710.5	
	T0408	5.9	
	T0421A	2.2	
	T0660A	29647	272003.6
Jeune Landing	6/200	-6.6	
	6/201	337.2	
	6/202	42202.1	
	6/203	32633.2	
Jeune Landing (cont'd)	6/205	1008.7	
	6/400	3601.2	

Operation	Mark	Volume	Total
	6/401	20872	
	6/402	2003.1	
	6/97	1479	
	T0508A	1337.9	
	T0514A	3986.7	
	T0514B	30104.5	
	T0514C	9701	
	T0514D	1518.3	
	T0526	25.7	
	T0526A	2830.9	
	T0526B	33831.7	
	T0526C	860.2	
	T0599	2161.3	190488.1
Mahatta River	6/400	16533.3	
	6/401	-5107.2	
	6/402	51120.5	
	T0599	54543.1	
	T0699		117089.7
Port Mcneill	25/4	153.5	
	25/400	0	
	25/402	0	
	25/403	0	
	25/404	2027.1	
	25/98	603.9	
	6/601	4891	
	6/602	908.2	
	6/98	4165.6	
	BZ002	276	
	HZ004	4434.9	
	L44327		
	T0191A	172.1	
	T0475A	2408.4	
	T0475B	10903.2	
	T0693A	1725.9	
	T0693B	7317.6	39987.4
Rupert Inlet	6/900	30058.9	
	6/901	3375	
	6/902	8836.6	
	T0481A	9743.2	52013.7
Winter Harbour	6/7	104.6	
	6/700	166	
	6/701	12577.1	
	6/702	29602.1	
	6/703	16064.6	
	HZ003	0	
	T0664A	13.4	
	T0664B	398.8	
	T0687	0	58926.6
Total Company Tenures			246823.9
Total Crown			621213.0
Grand Total			868036.9
Total Company and Phase Contractor Operations			600107.5
Total Harvest under Full Contracts			267929.4

Appendix II

TREE FARM LICENCE 6
1998 Volume Charged to Allowable Cut

CUBIC METRES

Mark	Crown Grant	Licences	Crown	TOTAL
25/4			153.5	
25/400				
25/402				
25/403				
25/404			2027.1	
25/98			603.9	
6/100			25.1	
6/102			7211.8	
6/104			108587.5	
6/200				
6/201			337.2	
6/202			42202.1	
6/203			32633.2	
6/205			1008.7	
6/300			243.4	
6/301			25248.4	
6/302			22284.0	
6/400			20134.5	
6/401			15764.8	
6/402			53123.6	
6/5				
6/600			5.6	
6/601			46597.2	
6/602			909.0	
6/61			334.9	
6/7			104.6	
6/700			166.0	
6/701			12577.1	
6/702			29602.1	
6/703			16064.6	
6/750			38739.6	
6/800			131.9	
6/801			8868.6	
6/802			12885.8	
6/803			28241.4	
6/804			35502.5	
6/870				
6/871				
6/872			408.5	
6/873			5607.0	
6/90			5113.8	
6/900			30058.9	
6/901			3375.0	
6/902			8836.6	
6/97			1479.0	
6/98			4165.6	
BZ002	276.0			
CZ010	9281.5			
HZ003				
HZ004		4434.9		
JZ004		43.7		
KZ003		108.0		
NAXNQ	1402.5			
T0191A		172.1		

Mark	Crown Grant	Licences	Crown	TOTAL
T0386A		4273.2		
T0386B		5720.6		
T0391A		2554.9		
T0399A		710.5		
T0408		5.9		
T0421A		2.2		
T0457A		0.8		
T0475A		2422.6		
T0475B		10903.2		
T0481A		9743.2		
T0508A		1337.9		
T0514A		3986.7		
T0514B		30104.5		
T0514C		9701.0		
T0514D		1518.3		
T0526		25.7		
T0526A		2830.9		
T0526B		33831.7		
T0526C		860.2		
T0599		56704.4		
T0660A		43817.9		
T0664A		13.4		
T0664B		398.8		
T0687				
T0693A		48.1		
T0693B		9474.5		
T0699				
T0700A		114.1		
Grand Total	10960.0	235863.9	621213.0	868036.9
Residue				
Recognized residue survey volumes associated with 1998 MOF S&R invoices				40970.9
Total Chargeable				909007.8

Appendix III

TREE FARM LICENCE 6 Area Denuded – 1998

HECTARES

Operation	Crown Grant	Crown / Licence	TOTAL	Medium, Low, and Very Low Ecosites
Holberg		234.3	234.3	76.0
Koprino		107.9	107.9	1.4
Michelson		53.9	53.9	0.0
Winter Harbour		63.3	63.3	12.5
Botel Lake		50.2	50.2	9.3
Jeune Landing		177.4	177.4	11.6
Mahatta		161.0	161.0	12.5
Coal Harbour		62.1	62.1	7.8
Rupert Inlet		30.0	30.0	2.9
Port McNeill	119.2	45.4	164.6	0.0
TOTAL	119.2	985.5	1104.7	134.0
Port McNeill Thinning		7.1	7.1	n/a
SBFEP		26.1	26.1	n/a

Appendix IV

TREE FARM LICENCE 6 Current Cut Control Period Annual Allowable Cut

CUBIC METRES

Year	Allowable Cut Available to Licencee	Chargeable Cut
1995	1 198 361	1 326 233
1996	1 275 506	1 325 989
1997	1 275 506	1 267 622
1998 ²	1 323 084	909 008
1999	1 464 264	
TOTAL	6 536 721	4 828 852 ¹

¹ The cumulative volume charged by the end of the fourth year of the cut control period is 4 828 852 m³ or 74 per cent of the five-year Allowable Cut.

² Volumes were amended to include TFL 25 Block 4 (Port McNeill) effective October 1, 1998.

Appendix V

TREE FARM LICENCE 6 Historical Cut Control Performance 1951 - 1998

CUBIC METRES

Cut Control Period	Allowable Cut Available to Licencee	Chargeable Cut
1951 – 1960	4 168 156	4 242 856
1961 – 1965	3 652 867	3 937 601
1966 – 1970	5 703 003	5 749 372
1971 – 1972	2 619 304	1 862 800
1973 – 1977	6 250 735	5 915 767
1978 – 1979	2 361 622	2 544 317
1980 – 1984	6 321 622	6 618 305
1985 – 1989	6 462 600	6 944 146
1990 – 1994	5 906 900	5 941 356
1995 - 1999	5 072 457	4 828 852
TOTAL	48 519 266	48 585 372

Volumes were amended to include TFL 25 Block 4 (Port McNeill) effective October 1, 1998.

Appendix VI

TREE FARM LICENCE 6
Coastal Contractor Clause Performance Report

CALENDAR YEAR 1998

Referenc	Description		Source
1)	Total AAC of TFL approved by Chief Forester (CF) that is available to Licencee	1 464 264 m ³	CF's approval letter for management and working plan
2)	AAC attributable to Schedule "B" lands that is available to Licencee	1 131 994 m ³	Derived from the approved MWP
3)	Volume of timber harvested	868 037 m ³	Obtained from the Regional Timber Officer of District Manager; the total volume of timber that is billed to the Licencee under the Licence during the calendar year (Section 49.1 of Forest Act)
4)	Harvested volume attributed to Schedule "B" lands	671 062 m ³	Calculated: (#2/#1) X #3
5)	Total volume contracted under full and phase contracts	470 710 m ³	Licencee Records
6)	Total volume contracted expressed as a per cent of compliance required	140 %	Calculated: (#5/(#4 X 0.5)) X 100

Licencee Name: Western Forest Products Limited

Completed by: William Dumont, R.P.F.

Date Report Completed: May 7, 1999

Appendix VII

TREE FARM LICENCE 6 Phase and Full Contractors - 1998

Contractor	Phase
A.H. Jackson Corp.	Full, Poling
C&E Road Building Ltd	Road
C.S.K. Hauling Ltd.	Hauling
Goodspeed Road Works Ltd.	Road
Helgesen Logging Ltd.	Hauling
Holberg Contracting Ltd.	Road
J.F. Trucking Ltd.	Road
Juthans Trucking Ltd.	Road
Koprino Contracting Limited	Road
Koprino Logging Contractors Limited	Full
	Road
Leeson Lake Contracting Ltd.	Falling
Lefty's Log Hauling	Hauling
MacLeod & Smith Log	Hauling
Mi Don Contracting Ltd.	Hauling
Poor Boy Enterprises Limited	Road
Port McNeill Enterprises Ltd.	Road
Pt. Alice Salvage Ltd.	Hauling
Quatsino Forest Company	Loading
	Yarding
Raft Cove Cont. Ltd.	Road
Ransom Creek Falling Ltd.	Falling
Stan Zwicker Contracting Ltd.	Road
Steele Rock Hauling Ltd.	Hauling
W.D. Moore Logging Co. Ltd.	Loading
	Road
	Yarding
G.L.M. Falling Limited	Falling
R. & K. Tyson Contracting Ltd	Hauling
D.D.H. Contracting Limited	Road
531088 B.C. Ltd.	Road
North Island RockPro Inc.	Road
557025 B.C. Ltd.	Full

Appendix VIII

TREE FARM LICENCE 6 Contractors - 1998

Name	Work	Location
4-Way Cedar	Salvage MP	Port McNeill
A.H. Jackson Corporation	Poling, Miscellaneous Culvert Hauling	Port McNeill
Andrea Anderson	Hatchery	Holberg
Angus Brown	Grounds Maintenance	Port McNeill
Arbor Tech	Terrain Assessments	Port McNeill
Associated Engineering	Engineering Consultant	Holberg
Azmeth Forest Consultants		Port McNeill
Builders Warehouse	Rigging deliveries	QDLS
Bruce Geotech Consultants	Terrain Assessments	Holberg
C & E Road Building Ltd.	Road construction	Jeune Landing
C. Fox	Tours	Port McNeill
Cathy Willson	Clerical	Holberg
Charcoal Creek	Research	Port McNeill
Coast Dryland Services Ltd.	Log Rehaul	Port McNeill
Coast Range Contracting	Silviculture - Planting	Holberg
Community Forestry Program	Spacing, Pruning	Port McNeill
CSK Hauling Ltd.	Log Hauling	Holberg
D. Anderson	Tours	Port McNeill
D. E. Whidden & Associates	Engineering Consultant	Holberg
D.D. & H Contracting	Road Construction	Holberg
D.K.Trucking & Contracting Ltd.	Road construction & deactivation	Holberg, Jeune Landing
Dabber Enterprises	Misc.backhoe work	QDLS
Danny Clarke	Fire Equipment Maintenance	Holberg
Dave Mallet	Carpentry	Holberg
Delaine Laviolette	Clerical	Holberg
DH Timber Towing & Salvage	Timber Towing	QDLS
Donna Larade	Grounds Maintenance, Janitor	QDLS,Jeune Landing
E. Skansen	Research	Port McNeill
Ed and Bonnie Cyr		Port McNeill
Evelyn McLaughlin	Janitor	Holberg
Fishfor Contracting Ltd.	Stream Sampling	Port McNeill, Jeune Landing, Holberg
Foresil Enterprises	Silviculture - Spraying	Jeune Landing, Holberg
Fortis Forestry Mgmt Services	Engineering Consultant	Holberg
French Creek Forest Services	Consulting services	Port McNeill, Holberg, Jeune Landing
G. Neal	Research	Port McNeill
G. Wickstrom	Tours	Port McNeill
G.L.M. Falling Ltd.	Falling & bucking	Port McNeill, Holberg, Jeune Landing
GCB Ventures Ltd.	Log Hauling	Holberg
GDS Contracting Ltd.	Yarding and Loading	Holberg
General Tree Planting Ltd.	Brushing	Jeune Landing
Golder Associates	Terrain Assessments	Holberg
Goodspeed Road Works Ltd.	Road Construction	Holberg
Helgesen Logging Ltd.	Log Hauling	Holberg
Herb Saunders Contraction	Road Construction	QDLS, Holberg
Holberg Contracting Ltd.	Road Construction - Drill	Holberg
Holberg Salvage and Disposal	Garbage Disposal	Holberg
I and I Forestry	Community Forestry	Port McNeill
I. R. Wilson	Archeological Surveys	Holberg
Interior Reforestation Co.Ltd.	Hydro & dry seeding	Port McNeill, Jeune Landing
J & T Silviculture	Spacing and Pruning	Port McNeill, Holberg
J.F.Trucking Ltd.	Road construction	Port McNeill, Holberg, QDLS, Jeune Landing
Jaco Powerlines Ltd.	Electrical Contractor	Holberg
JRP consulting	Development Plan (Mapping)	Port McNeill

Name	Work	Location
Juthans Trucking Ltd	Road Construction	Holberg
Kellas & Co. Forest Consulting	Engineering consultant	Jeune Landing
Kelsey For. Eng. Service	Engineering consultant	Jeune Landing
Kindry Louie	Firewood Contract	Holberg
Koprino Logging Contractors	Log hauling, Road Construction	Holberg, Jeune Landing
Leader Silviculture	Spacing/Pruning	Jeune Landing
Leanne Boas	Janitorial services	Jeune Landing
Leeson Lake Logging Ltd.	Falling and Bucking	Holberg
Lefty's Log Hauling Ltd.	Log hauling	Jeune Landing
Lemare Lake Logging Ltd.	Logging contractor	Jeune Landing
Long Line Logging Ltd.	Yarding and Loading	Holberg
Lorna Irving	Clerical	Holberg
M.Jones Enterprises	Pit work (capping, etc.)	QDLS
Macleod & Smith Log Hauling	Log hauling	Jeune Landing
Mahatta Cedar Products Ltd.	Cedar salvage	Port McNeill, Jeune Landing
Mark Bannerman	Firewood Contract	Holberg
Midon Trucking Ltd.	Log Hauling	Holberg
Mike Jones Enterprises Ltd.	Backhoe work	Jeune Landing
Morris Haugrud	Machine Cleaning	Holberg
Ned Boniface	Parks Maintenance, Firewood	Port McNeill
North Island Diving	Carriage repairs	QDLS
North Island Power Wash	Steam cleaning	QDLS
North Island Rockpro Inc	Road Construction	Port McNeill
Nu Creek Developments	Excavating	Jeune Landing
O.K.Paving	Resurfacing sort	QDLS
Otter Enterprises	Equipment/misc.repairs	QDLS
Pacific Forest Maintenance	Pruning, Spacing	Jeune Landing, Port McNeill
Pacific Rainforest Scaling	Scaling	QDLS
Piteau & Associates Ltd.	Terrain Assessments	Port McNeill, Holberg
Poor Boy Enterprises	Backhoe	Port McNeill
Port Alice Dredging	Dredging	QDLS
Port Alice Logging	Rebundling WFP boomsticks	QDLS
Port Alice Salvage Ltd.	Log hauling	Jeune Landing
Port Hardy Bulldozing	Road Construction	Holberg
Port McNeill Consulting		Port McNeill
Port McNeill Enterprises	Backhoe	Port McNeill
Port McNeill Shake and Shingle	Salvage	Port McNeill
Practical Tree Improvement	Research	Port McNeill
Quatsino Cablevision Ltd.	Electrician	Holberg
Quatsino Forest Company	Silviculture	Port McNeill
R & K Tyson Contraction Ltd.	Log Hauling	Holberg
R.J.Hay Consulting Ltd.	Engineering consultant	Jeune Landing
Race Enterprises	General labour	Jeune Landing
Raft Cove Contracting	Road Construction	Holberg
Ransom Creek Falling Ltd.	Falling & bucking	Jeune Landing
Rare Image	Research	Port McNeill
Rebecca Willson	Hatchery	Holberg
Riverside Bridge & Pile Ltd.	Road construction	Holberg, Jeune Landing
Roosevelt Cedar	Salvage	Port McNeill
Schaefer Contracting Ltd.	Yarding and Loading	Holberg
Shawn Hamilton	Inventory	Port McNeill
Simons Reid Collins	Timber Inventory	Port McNeill
Sources Archaeology	Archaeological Inventory	Holberg
Stan Zwicker Contracting Ltd	Backhoe, Lowbed	Port McNeill
Steel Rock Contracting Ltd.	Log Hauling	Holberg
Sumas Valley Enterprises	Silviculture - Alder Control	Holberg
Sun West Helicopters	Aerial Pruning	Holberg
T. Lewis	Training	Port McNeill
Terra Engineering	Terrain Assessments	Holberg
Terra Firma	Terrain Assessments	Jeune Landing
Terra Tech	Terrain Assessments	Port McNeill
Thurber	Terrain Assessments	Jeune Landing

Name	Work	Location
Tim Smith	Terrain Assessments	Holberg
Timberline Reforestation Ltd.	Tree planting	Port McNeill, Jeune Landing
TMR Enterprises	Engineering Consultant	Holberg
Tribune Timber Ltd.	Stump to dump contractor	Port McNeill, Jeune Landing
W. D. Moore Logging	Road Construction	Holberg
Wallas Forestry	Pile Burning, Spacing, Pruning	Holberg, Port McNeill, Jeune Landing
Wallas Forestry	Spacing/Pruning	Port McNeill
Warner Bay Investments Ltd.	Yarding and Loading	Holberg
Westrek		
Western Aerial Applications Ltd.	Aerial foliar spray	Jeune Landing
Willdon Enterprises	Building maintenance	Holberg, QDLS

Appendix IX

TREE FARM LICENCE 6 Partitioned Harvest Performance Summary - 1998

MEDIUM, LOW, AND VERY LOW SITES¹

Operation	M/L/VL Ecosites ²	
	Area (ha)	Volume (m ³)
Holberg	76.0	69 464
Koprino	1.4	1 280
Michelson	0.0	0
Winter Harbour	12.5	11 425
Botel Lake	9.3	8 500
Jeune Landing	11.6	10 602
Mahatta	12.5	11 425
Coal Harbour	7.8	7 129
Rupert Inlet	2.9	2 651
Port McNeill	0.0	0
TOTAL	134.0	122 476

¹ Performance must be 52 000 m³ minimum per year on a five-year average.

² Inventory/MP basis only. Volume estimates per hectare are higher than actual production.

Helicopter

There was no helicopter logging in TFL 6 in 1998.

Appendix X

TREE FARM LICENCE 6
Small Business Forest Enterprise Program
Harvesting Report – 1998

Year	Volume Available to SBFEP m ³	Licence No.	Licensee	Area Logged ha	Volume Scaled m ³	Residue m ³	Chargeable Volume m ³	TS Status
1988	25800			0	0	0	0	
1989	61673	TS24141	Koskimo Cont.	3.4	0	0	0	
1990	100620	TS24141	Koskimo Cont.	0	2733	256	2989	Complete
		TS35673	Primex	27.0	9649.0	0	9649	
1991	100620	TS35673	Primex	65.4	62813.0	0	62813.0	
1992	100620	TS35673	Primex	101.3	84958.0	0	84958.0	
		TS37073	Hudson Mitchell	0	0	0	0	
1993	100620	TS35673	Primex	114.1	72571.0	6952.0	79523.0	
		TS37073	Hudson Mitchell	54.9	74016.0	1127.0	75143.0	
		TS39536	BW Creative Wood	0	0	0	0	
1994	100620	TS35673	Primex	65.3	62274.0	2209.0	64483.0	
		TS37073	Hudson Mitchell	48.0	23959.0	0	23959.0	
		TS39536	BW Creative Wood	0	0	0	0	
1995	100620	TS35673	Primex	0	594.0	0	594.0	Complete
		TS37073	Hudson Mitchell	88.0	51075.0	1396.0	52471.0	
1996	0	TS37073	Hudson Mitchell	91	60985.0	985.0	61970.0	
1997	0	TS37073	Hudson Mitchell	0	4412.0	5328.0	9740.0	
1998	13242	TSL 43355	BW Creative Wood	25.3	22526.0	0	22526.0	
TOTAL	704435			683.7	532565	18253	550818	

Appendix XI

Project Summary - 1998

Project			Port McNeill	Holberg	Jeune Landing	TOTAL
Denudation	Total Company	ha	256.7	509.6	338.4	1 104.7
	Crown / Licence	ha	137.5	509.6	338.4	985.5
	Crown Grant (MF)	ha	119.2			119.2
	SBFEP	ha	25.3			25.3
	TOTAL	ha	282.0	509.6	338.4	1 130.0
Accidental Fires			No.	3	0	3
			ha	5.9		5.9
Site Preparation	Prescribed Burning	ha				
	Pile Burning	km	24.1	58.6	6.5	89.2
	Mechanical	ha	23.6	2.0		25.6
	Crown / Licence	ha	23.6	2.0		25.6
	Crown Grant (MF)	ha				
Planting	Original	ha	523.2	554.8	255.7	1333.7
	Replants	ha	13.2	2.2		15.4
	Fill-ins	ha	1.2	14.4	24.3	39.9
	Total	ha	537.6	571.4	280.0	1389.0
	Crown / Licence	ha	438.4	550.7	280.0	1269.1
	Crown Grant (MF)	ha	99.2	20.7		119.9
	Number of Seedlings	Cw	280796	207569	52910	541275
		Hw	155636	138545	46519	340700
		Ba	110532	178572	133053	422157
		Fdc	9755	4400	13490	27645
		Yc	54474	2448	16022	72944
		Ss	8720	50214	2160	61094
		P	900	4380	2100	7380
		Misc		400		400
		Total	620813	586528	266254	1473595
Stocking Surveys	Natural Regeneration	ha	71.1	9.4	109.5	190.0
	Sufficiently restocked	ha	63.7	9.4	106.9	180.0
	Crown / Licence	ha	31.9	9.4	106.9	148.2
	Crown Grant (MF)	ha	31.8			31.8
	Other Classification	ha			2.6	2.6
	Crown / Licence	ha			2.6	2.6
	Crown Grant (MF)	ha				
	Plantation Regeneration	ha	1092.6	1022.0	243.7	2358.3
	Sufficiently Restocked	ha	1065.3	1012.0	236.8	2314.1
	Crown / Licence	ha	1026.1	1011.8	236.8	2274.7
	Crown Grant (MF)	ha	38.3	0.2		38.5
	Not Sufficiently Restocked	ha	27.3	10.0	5.5	42.8
	Crown / Licence	ha	23.4	10.0	5.5	38.9
	Crown Grant (MF)	ha	3.9			3.9
	Other Classification	ha			1.4	1.4
	Crown / Licence	ha			1.4	1.4
	Crown Grant (MF)	ha				
Silviculture Prescriptions			ha	697.8	1034.4	2200.8
			No.	18	31	65
Plantation Survival Assessments			ha	560.3	724.8	1444.1
			%	92.8	95	91

Project			Port McNeill	Holberg	Jeune Landing	TOTAL
Free Growing Surveys	Total	ha	4141.6	2776.9	4386.0	11304.5
	Free Growing	ha	2859.3	1023.7	2625.0	6508.0
	Crown / Licence	ha	2470.7	1023.7	2537.0	6031.4
	Crown Grant (MF)	ha	388.6		88.0	476.6
	Not Free Growing	ha	1258.4	1753.2	1761.0	4772.6
	Crown / Licence	ha	1147.8	1752.0	1676.0	4575.8
	Crown Grant (MF)	ha	110.6	1.2	85.0	196.8
	Other Classification	ha	23.9			23.9
	Crown / Licence	ha	21.8			21.8
	Crown Grant	ha	2.1			2.1
Brushing and Weeding	Total	ha	27.2	656.2	390.6	1074.0
	Manual	ha	27.2		39.4	66.6
	Mechanical	ha		430.6		430.6
	Stem Injection	ha		213.3	305.5	518.8
	Aerial Foliar	ha			16.0	16.0
	Ground Foliar	ha		12.3	23.2	35.5
	Crown / Licence	ha	13.1	652.9	390.6	1056.6
	Crown Grant (MF)	ha	14.1	3.3		17.4
Juvenile Spacing	Total	ha	15.6	37.9	22.9	76.4
	Crown / Licence	ha	15.6	37.9	22.9	76.4
	Crown Grant (MF)	ha				
Pruning	Total	ha	186.9	39.5	30.2	256.6
	Crown / Licence	ha	186.9	39.5	30.2	256.6
	Crown Grant (MF)	ha				
Fertilization	Total	ha	0.0	0.0	0.0	0.0
Commercial Thinning	Total	ha	7.1			7.1
	Crown / Licence	ha	7.1			7.1
	Crown Grant (MF)	ha				
Residue Assessment Plots			38	223	64	325
CP Cruising Plots						997
Minor Products	Cedar Shake and Shingle	m ³	5259.5	5332.7	2164.0	12756.2
	Yew Bark	kg	1580.0		1352.0	2932.0
	Gravel and Sand	m ³	31573.0			31573.0
	Honey	kg	17455.0		680.0	18135.0
	Cedar Oil	l	512.0			512.0
	Salvage Logs	m ³	829.3		403.0	1232.3
	Cedar Cants	m ³		63.9		63.9
	Guitar Blocks	m ³		9.1		9.1
Engineering	Roads Constructed	km	14.4	31.2	20.9	66.5
	Roads Re-Built	km	0.6	13.9	2.8	17.3
	Roads Maintained	km	241.9	214.0	312.0	767.9
	Roads Deactivated					
	Seasonal	km	0.0	11.3	9.0	20.3
	Semi-Permanent	km	11.4	4.3	3.8	19.5
	Permanent	km	2.7	6.8	4.1	13.6
Roadside Treatments	Mechanical Brushing	km	0	36.8	13	49.8
	Chemical Spraying	km	0	0	6.5	6.5
	Hydro and Dry Seeding	km	40.3	0.4	58.4	99.1
Site Stabilization			3.5	14.6	7.5	25.6
Salmonid Enhancement Program	Total		241805	90397	67500	399702
	Chinook		201379	3483	7000	211862
	Coho		37226	66330	60500	164056
	Chum			20584		20584
	Steelhead		3200			3200

Appendix XII

**Western Forest Products
Regeneration Balance Sheet
to December 31, 1998**

Item	TFL 6	TFL 19	TFL 25	Total
OPENING BALANCE (NSR at 1998 01 01)	1951 ha	1037 ha	2262 ha	5250 ha
Debits				
1998 Denudations	1132 ha	631 ha	448 ha	2211 ha
1998 Surveys	43 ha	258 ha	106 ha	407 ha
Credits				
1998 Planting	1341 ha	738 ha	651 ha	2730 ha
1998 Surveys (Natural Regeneration)	180 ha	0 ha	14 ha	194 ha
Other	4 ha	0 ha	35 ha	39 ha
CLOSING BALANCE (NSR at 1998 12 31)	1601 ha	1188 ha	2116 ha	4905 ha

Appendix XIII

TREE FARM LICENCE 6 Historical Summary of Activities										
Year	Denuded (ha)	Planted (ha)	No. Trees Planted	Juvenile Spacing (ha)	Brushing (ha)	Prescribed Burning (ha)	Mechanical Site Prep. (ha)	Fertilization (ha)	Pruning (ha)	Commercial Thinning (ha)
Pre										
1965	13793.2	797.7	701400	0.0	174.2	484.9	51.4	0.0	0.0	0.0
1965	1517.5	497.2	361500	20.3	0.0	1514.8	0.0	0.0	0.0	0.0
1966	1756.5	430.9	325300	30.4	0.0	1382.0	0.0	0.0	0.0	0.0
1967	1744.8	674.9	422950	16.2	12.2	1654.2	57.0	0.0	0.0	0.0
1968	1780.8	620.0	444900	0.2	54.6	1363.8	0.0	0.0	0.0	0.0
1969	1742.7	1126.1	989650	0.8	0.0	1195.6	0.0	0.0	0.0	0.0
1970	1469.4	1155.0	751700	17.1	0.0	1177.9	14.6	0.0	0.0	0.0
1971	1751.7	717.8	529350	83.0	30.0	593.2	5.7	0.0	0.0	0.0
1972	1310.2	1066.2	912650	136.1	0.0	1175.1	24.7	0.0	0.0	0.0
1973	1660.7	569.6	600500	105.7	6.1	790.0	8.5	0.0	0.0	0.0
1974	1496.5	459.1	459350	37.9	112.6	646.5	5.7	0.0	0.0	0.0
1975	944.5	911.5	777700	2.2	276.2	10.9	4.5	0.0	0.0	0.0
1976	1875.1	983.6	777050	557.7	390.8	1439.6	0.0	0.0	0.0	0.0
1977	2137.4	1047.8	553900	729.3	726.6	1423.8	0.0	3.5	0.0	0.0
1978	1977.2	692.3	493950	720.4	72.9	23.1	0.0	0.0	0.0	0.0
1979	1879.8	837.4	662850	566.5	127.9	1090.0	1.6	0.0	0.0	0.0
1980	1686.4	566.8	491500	462.0	342.3	374.0	0.0	0.0	0.0	0.0
1981	1378.4	1213.7	1047600	333.4	279.8	231.1	0.0	0.0	0.0	40.9
1982	1659.2	1256.7	1198300	172.2	0.0	335.4	0.0	0.0	0.0	0.0
1983	2313.3	936.6	888000	0.0	373.7	571.7	45.3	0.0	0.0	0.0
1984	2315.2	960.2	882400	132.7	1768.9	421.0	24.0	0.0	26.9	0.0
1985	2234.1	648.2	701800	485.6	712.1	1050.8	118.4	0.0	17.3	0.0
1986	1772.1	1301.1	1347100	303.3	1099.9	1526.2	59.2	200.0	9.0	0.0
1987	2284.8	2258.4	2256650	419.6	2668.4	967.4	115.6	0.0	9.1	0.0
1988	1849.5	1866.9	1844050	65.4	899.7	1008.9	11.1	0.0	2.0	0.0
1989	1173.4	1118.8	1169250	469.9	2053.4	874.7	64.7	838.0	5.5	0.0
1990	1394.3	1382.0	1405700	548.0	1511.7	617.0	35.2	795.0	8.4	0.0
1991	1642.6	1801.9	1491100	827.1	1656.2	347.5	50.2	0.0	63.8	0.0
1992	1467.2	1522.5	1550900	377.1	1429.2	491.7	14.2	0.0	34.9	0.0
1993	1881.7	1524.1	1574650	97.5	1357.5	499.7	86.9	0.0	42.4	0.0
1994	1666.8	1697.2	1712150	829.1	843.9	424.2	92.4	0.0	178.5	0.0
1995	1661.7	2060.6	2003400	802.6	1487.1	76.6	47.7	516.1	350.4	7.5
1996	1697.5	2065.8	2110950	549.7	1475.7	267.8	87.8	1936.2	659.9	0.0
1997	1404.2	1789.4	1944750	539.5	1066.3	175.3	24.3	2323.3	916.5	0.0
1998	1104.7	1333.7	1473600	76.4	1074.0	0.0	25.6	0.0	256.6	7.1
TOTAL	71425.1	39891.7	36858550	10514.9	24083.9	26226.4	1076.3	6612.1	2581.2	55.5

- Historical Summary amended to include all of the former TFL 25 Block 4.

Appendix XIV

Western Forest Products Tree Planting History

Number of Seedlings						
Year	TFL 6	TFL 19	TFL 25	WFP Misc. Properties	Other Properties	TOTAL
Pre 1965	701400	3502000	2531100	3782450		10516950
1965	361500	425000	298500	247700		1332700
1966	325300	726000	432800	0		1484100
1967	422950	434000	547650	285800		1690400
1968	444900	539000	645250	46700		1675850
1969	989650	474000	446100	327300		2237050
1970	751700	535000	341450	136500		1764650
1971	529350	1123000	586700	158250		2397300
1972	912650	912000	295300	407000		2526950
1973	600500	699000	772450	162000		2233950
1974	459350	1324000	363850	57950		2205150
1975	777700	942000	199450	67950		1987100
1976	777050	709000	807250	58250		2351550
1977	553900	631000	757550	172350		2114800
1978	493950	494000	555600	38800		1582350
1979	662850	524000	749000	12300		1948150
1980	491500	473000	493650	24150		1482300
1981	1047600	579000	803900	29900		2460400
1982	1198300	735000	827700	16900		2777900
1983	888000	566000	669050	55450		2178500
1984	882400	325000	809000	102700		2119100
1985	701800	452000	522050	69550		1745400
1986	1347100	346000	630950	57000		2381050
1987	2256650	686000	1297750	329300		4569700
1988	1844050	563000	982850	172950		3562850
1989	1169250	755000	735600	287750		2947600
1990	1405700	707000	712350	354150		3179200
1991	1491100	439000	842850	60150		2833100
Pre 1992					5268800	5268800
1992	1550900	757000	673900	95700	232850	3310350
1993	1574650	683000	639750	240600	377550	3515550
1994	1712150	674000	546000	226200	790600	3948950
1995	2003400	1040000	853050	160400	894500	4951350
1996	2110950	1140000	1089750	579400	3550	4923650
1997	1944750	1067000	951900	52150	0	4015800
1998	1473600	675550	652250	1522400	28750	5826150
TOTAL	36858550	26655550	25064300	10398100	7596600	108046700

- TFL 6 Summary amended to include the former TFL 25 Block 4
- TFL 25 Summary amended to include the former TFL 24, and exclude the former TFL 25 Block 4

Appendix XV – A

Direct Employment Summary – TFL 6

Operation	Home Region	Contractor Personnel		Company Personnel		TOTAL	
		People	Person Days	People	Person Days	People	Person Days
Planning, Engineering, and Road Development							
Head Office	South Mainland	4	134	4	334	8	468
Region	North Vancouver Island	16	712	1	220	17	932
Port McNeill	North Vancouver Island	14	1039	6	1400	20	2439
Jeune Landing	North Vancouver Island	17	2790	6	1046	23	3836
	Vancouver Island	2	160	0	0	2	160
	South Mainland	1	80	0	0	1	80
Holberg	North Vancouver Island	33	2352	7	1524	40	3876
	Vancouver Island	10	964	3	682	13	1646
	South Mainland	7	272	0	0	7	272
SUBTOTAL		104	8503	27	5206	131	13709
Harvesting							
Port McNeill	North Vancouver Island	21	4409	64	11379	85	15788
Quatsino Dryland Sort	North Vancouver Island	66	1849	33	5597	99	7446
	Vancouver Island	0	0	1	189	1	189
Jeune Landing	North Vancouver Island	74	6628	50	6909	124	13537
	Vancouver Island	2	246	1	131	3	377
Holberg	North Vancouver Island	99	15040	58	5361	157	20401
	Vancouver Island	8	1100	7	977	15	2077
	South Mainland	2	215	1	130	3	345
SUBTOTAL		272	29487	215	30673	487	60160
Silviculture and Integrated Resource Management							
Saanich Forestry Centre	Vancouver Island	0	0	46	1927	46	1927
Region	North Vancouver Island	18	691	0	0	18	691
Port McNeill	North Vancouver Island	23	1502	5	1200	28	2702
	Vancouver Island	20	897	0	0	20	897
	South Mainland	8	120	0	0	8	120
Jeune Landing	North Vancouver Island	48	507	12	1454	60	1961
	Vancouver Island	29	632	0	0	29	632
	South Mainland	14	289	0	0	14	289
Holberg	North Vancouver Island	39	468	10	1741	49	2209
	Vancouver Island	21	243	1	84	22	327
	South Mainland	86	1622	0	0	86	1622
SUBTOTAL		306	6971	74	6406	380	13377

Operation	Home Region	Contractor Personnel		Company Personnel		TOTAL	
		People	Person Days	People	Person Days	People	Person Days
Transportation							
Vancouver	South Mainland	79	7071			79	7071
SUBTOTAL		79	7071			79	7071

Processing							
Port Alice Pulp Mill	North Vancouver Island			495	23928	495	23928
Squamish Pulp Mill	South Mainland			347	12620	347	12620
Duke Point Sawmill	Vancouver Island			88	18810	88	18810
Ladysmith Sawmill	Vancouver Island			100	6216	100	6216
Cowichan Bay Sawmill	Vancouver Island			87	1091	87	1091
Silvertree Sawmill	South Mainland			133	6699	133	6699
Vancouver Sawmill	South Mainland			143	2696	143	2696
Nanaimo Sawmill	Vancouver Island			140	2009	140	2009
Nanaimo Log Merchandising	Vancouver Island			40	391	40	391
Log Trades and Sales	South Mainland			35	1168	35	1168
SUBTOTAL				1608	75628	1608	75628

Administration							
Head Office	South Mainland			34	2842	34	2842
Region	North Vancouver Island	2	70	6	1320	8	1390
SUBTOTAL		2	70	40	4162	42	4232

Summary – By Home Region							
	North Vancouver Island	470	38057	753	63079	1223	101136
	Vancouver Island (Other)	92	4242	514	32507	606	36749
	South Mainland	201	9803	697	26489	898	36292
TOTAL		763	52102	1964	122075	2727	174177

Appendix XV - B

**Western Forest Products
Direct Employment Statistics
(person-days*)**

	Planning, Engineering, and Road Development	Harvesting	Transportation	Processing	Silviculture and Integrated Resources Management	Administration	TOTAL
TFL 6	13709	60160	7071	75628	13377	4232	174177
TFL 19	9147	38287	3536	51309	5918	4023	112220
TFL 25	7723	16010	2232	32461	7992	1280	67698
MF 61	0	0	0	0	147	0	147
FL A16845	2181	6194	**	12574	1682	611	23242
FL A16847	2908	6419	**	14918	1418	677	26340
FL A19205	683	1487	**	2908	186	132	5396
FL A19216	1804	5655	**	8287	304	316	16366
FL A19228	128	1772	**	3819	77	52	5848
FL A19231	4210	24373	**	**	1261	2269	32113
FL A19240	11	0	**	32724	1122	84	33941
FL A53746	236	910	**	**	170	214	1530
Other Tenures	0	0	5768	15250	1531	0	22549
Campbell Island	0	0	0	0	571	0	571
TOTAL	42740	161267	18607	249878	35756	13890	499874

* Includes Company and Contract Personnel.

** See Other Tenures

Appendix XV - C

Western Forest Products
First Nations Silviculture Contracts Employment Summary
1998

	TFL 6	TFL 19	TFL 25	Campbell Island	FL A16845	FL A19228	FL A53746 Zeballos Community	FL A19231	FL A19240	MF 61	Total	Total Contractor Days	Percent First Nations Employment	Percent WFP Goal
Holberg	475										475	2333	20	100
Jeune Landing	270										270	1428	19	95
Port McNeill	1242		2								1244	2519	49	245
Central Coast			477	491	50						1018	3201	32	160
Mainlands Islands			954			12					966	4037	24	120
Zeballos		32					170	592			794	1148	69	345
Gold River		514									514	1663	31	155
Nootka Contract Admin		0									0	678	0	0.0
TOTAL											528	17007	3	15

Tree Farm Licence 6									
Salmonid Enhancement Program Summary									
No. Fry Released (000)									
Year	Hatchery	Release Stream	Coho	Chum	Chinook*	Steelhead	Pink	Subtotal	TOTAL
1998	Marble	Marble River	37.2	0	201.4	3.2		241.8	399.7
	Colonial	Colonial Creek	60.5		7.0			67.5	
	Cordy	Goodspeed River San Josef River	23.9 42.4	20.6	3.5			48.0 42.4	
1997	Marble	Marble River	44.7		741.3			786.0	1045.6
	Colonial	Colonial Creek	19.7		56.7	9.7		86.1	
	Cordy	Goodspeed River Holberg Inlet San Josef River	10.9 49.2	65.4	29.1 18.9			105.4 18.9 49.2	
1996	Marble	Marble River	110.7		127.2			237.9	462.3
	Colonial	Colonial Creek	65.8		12.8			78.6	
	Cordy	Goodspeed River San Josef River Pegettem River	32.9 50.4	33.6	19.6			86.1 50.4 9.3	
					9.3				
1995	Marble	Marble River	40.0		300.0	6.0		346.0	727.0
	Colonial	Colonial Creek	80.1		38.2			118.3	
	Cordy	Goodspeed River* San Josef River	95.0 8.3	141.9	17.1	0.4		254.4 8.3	
1994	Marble	Marble River	42.0		700.0	6.0		748.0	983.5
	Colonial	Colonial Creek*	80.0		36.8	10.9		127.7	
	Cordy	Goodspeed River* San Josef River	13.9 31.0	34.4	21.0	7.5		76.8 31.0	
1993	Marble	Marble River	42.0		74.0	8.0		124.0	266.4
	Colonial	Colonial Creek	19.5					19.5	
	Cordy	Goodspeed River San Josef River	20.1 33.0	62.0		7.8		89.9 33.0	
1992	Marble	Marble River	78.0		312.2			390.2	732.2
	Colonial	Colonial Creek*	109.7		19.1			128.8	
	Cordy	Goodspeed River* San Josef River	98.7 1.0	82.8	21.5	7.0	2.2	212.2 1.0	
1991	Marble	Marble River	49.3		65.0	7.0		121.3	359.6
	Colonial	Colonial Creek	100.0					100.0	
	Cordy	Goodspeed River San Josef River	76.2 25.4	42.4		7.7	11.0	137.3 1.0	
1990	Marble	Marble River	54.3		629.8	9.0		693.1	977.4
	Colonial	Colonial Creek*	44.0		26.5	14.4		84.9	
	Cordy	Goodspeed River*	92.4	77.0	30.0			199.4	
1989	Marble	Marble River	24.0	0	120.0	1.4		145.4	319.2
	Colonial	Colonial Creek*	27.0	0	6.3			33.3	
	Cordy	Goodspeed River*	27.0	65.0	30.5	9.0		131.5	
1988	Marble	Marble River Waukwass	49.6		350.0	7.7		399.6 7.7	748.4
	Colonial	Colonial Creek*	80.0		8.0			88.0	
	Cordy	Goodspeed River*	15.5	102.0	8.8	3.3		129.6	
	Monkey	Mahatta River Monkey Creek	95.0 9.5	19.0				95.0 28.5	
to 1987	Marble	Marble River	122.0	2.8	1,243.7			1368.5	1928.5
	Cordy	Goodspeed	132.0	71.0				203.0	
	Colonial	Colonial Creek	69.0			3.0		72.0	
	Nequiltpaalis	Nequiltpaalis		42.0				42.0	
	Monkey	Mahatta River Monkey Creek	123.0 33.0	87.0				123.0 120.0	
TOTAL								8949.8	

Appendix XVII

WESTERN FOREST – WESTERN PULP – DOMAN INDUSTRIES – DOMAN WESTERN LUMBER

1998 Log Flow and Wood Consumption
(Approximate)

SAWMILL PULPMILL	TENURE / SOURCE (THOUSAND CUBIC METRES)											Inventory / Purchase	Total Consumption	Chips to Woodfibre (Thousand Units)
	TFL 6	TFL 19	TFL 25	FL A19240 Strathcona	FL A19240 Kingcome	FL A16845 Mid Coast	FL A16847 Mid Coast	FL A19205 Fraser	FL A19228 Sunshine	FL A19216 Soo	Other Tenures			
Duke Point	259											15	274	34
Chemainus												130	130	9
Ladysmith	111	10	79	5	7	25			1	2	9	146	395	22
Cowichan	23	24	55	16	2	8	3	1	1	4	24	214	375	25
Silvertree	80	80	81	54	6	11	13	1		2	26	1	355	51
Vancouver	44	12	100	6	19	37					13	102	333	46
Tahsis		154		80							8	62	304	10
Saltair		7		1			29	8	14	31	5	355	450	24
Nanaimo	33	24	4	1		3	56	12	16	27	12	315	503	10
Log Merchandizer	31	13	54	6	2	8		1	2	2	10	218	347	160
Port Alice Pulp Mill	176		21		11	16					5	346	575	
Trades / Sales	111	201	86	102	9	16	43	6	2	13	28	115	732	(7)
Squamish Pulp Mill Chips Purchased														219
Consumed														608
TOTAL LOGS	868	525	480	271	56	124	144	29	36	81	140	2019	4773	

Appendix XVIII

Forest Research Summary Permanent Plots and Trials

Salal Cedar Hemlock Integrated Research Program (Fertilization and Scarification Trials)

Trial (Year Established)	Location	Measurements	Last Report	Other
SCHIRP Establishment Trial (1988)	Port McNeill	1997, 1994, 1990, 1988, 1987	1996 Update	Foliar Sampling (1997)
Sitka Spruce Demonstration Trial (1984)	Port McNeill	1998, 1996, 1990, 1989, 1988, 1987, 1986, 1985, 1984	1994	Foliar Sampling (1997) Re-fertilize (1997)
Western Red Cedar Demonstration Trial (1987)	Port McNeill	1998, 1996, 1994, 1993, 1992, 1991, 1990, 1989, 1988, 1987, 1986	1994	Foliar Sampling (1997) Re-fertilize (1997)
Western Hemlock Demonstration Trial (1987)	Port McNeill	1998, 1996, 1994, 1993, 1992, 1991, 1990, 1989, 1988, 1987, 1986	1994	Foliar Sampling (1997) Re-fertilize (1997)
Salal Eradication Trial (1984)	Port McNeill	1994	1996 Update	
S1CH Scarification Trial (1996)	Port McNeill	1997, 1995	1996 Est Rep	Salal Measures (1998)
S1CH Individual Tree Fertilization Trial (1996)	Port McNeill	1997, 1996, 1995	1996 Est Rep	
Individual Tree Fertilization (1996)	Holberg	1997, 1995	1996 Est Rep	
Transitional S1CH/S1HA Trials (1996)	Holberg Port McNeill	1997, 1995	1996 Est Rep	Salal Measures (1998)
Salal Fertilization Trial (1993)	Port McNeill	1993	1994	
Operational Hand Fertilization (1996)	Port McNeill Holberg	1997, 1995	1996 Est Rep	
Operational Fertilization Monitoring (1997)	Port McNeill Holberg	1996		
Operational Fertilization Monitoring (1996)	Port McNeill	1997, 1995		
Operational Fertilization Monitoring (1995)	Port McNeill	1996, 1994		
Operational Fertilization Monitoring (1990)	Port McNeill Holberg Jeune Landing	1995, 1990, 1989	1997 Interim Report	
Operational Fertilization Monitoring (1989)	Port McNeill Holberg	1994, 1990, 1989	1997 Interim Report	
Operational Fertilization Monitoring (1986)	Port McNeill	1991, 1987, 1986	1994	
Organic Fertilizer Trials (1990)	Port McNeill	1995, 1993, 1991	1996 Update	
Fish Silage Trial (1993)	Port McNeill	1994, 1993	1996 Update	
Straw and Fish Compost (1994)	Port McNeill	1995, 1994, 1993, 1992	1996 Update	

Other Silviculture Research

Trial (Year Established)	Location	Measurements	Last Report	Other
<i>Suquash Drainage Trial (1997)</i>	Port McNeill	1998, 1997	1998 Est Rep	
<i>Western Red Cedar Stock Type Trial (1996)</i>	Port McNeill	1997, 1996	1997 Est Rep	
<i>S4 Trial (1984)</i>	Holberg	1995, 1990, 1988, 1987, 1986, 1985, 1984	1989 5-year Results	2 nd pruning Pruning assessments
<i>Cottonwood Nurse Tree Trial (1994)</i>	Jeune Landing		1996 Est Rep	
<i>Weevil Resistance Trial (1997)</i>	Jeune Landing		1997 Est Rep	

Genetics Trials

Trial (Year Established)	Location	Measurements	Last Report	Other
<i>Yellow Cypress Pilot (1988)</i>	Port McNeill	1996, 1995, 1994, 1993, 1991, 1990, 1989, 1988		Mortality Checks – 1997, 1996
<i>Yellow Cypress Phase 1 (1991)</i>	Port McNeill Jeune Landing	1997, 1994	1996 Interim Rep	
<i>Yellow Cypress Phase 2 (1992)</i>	Port McNeill	1998, 1995		
<i>Yellow Cypress Phase 3 (1993)</i>	Port McNeill Jeune Landing	1996		
<i>Yellow Cypress Phase 4 (1993)</i>	Port McNeill	1996		
<i>Yellow Cypress Phase 5 (1994)</i>	Port McNeill	1997		
<i>Yellow Cypress Phase 6 (1995)</i>	Port McNeill	1998		
<i>Yellow Cypress Phase 7 (1995)</i>	Holberg			
<i>Yellow Cypress Phase 8 (1995)</i>	Port McNeill			
<i>Coastal Douglas-fir Progeny Trial (1993)</i>	Port McNeill	1997	1993 Est Rep	
<i>Hybrid Poplar Clonal Trial (1991)</i>	Holberg	1993, 1991	1991 Est Rep	
<i>Sitka Spruce Clonal Trial (1988)</i>	Holberg		1988 Est Rep	

Growth and Yield Monitoring

Trial (Year Established)	Location	Measurements	Last Report	Other
<i>Type III Growth and Yield Installations (1988)</i>	Port McNeill Holberg	1995	1995 MOF Report	Vegetation Measures - 1997
<i>Growth And Yield Plots (1994)</i>	Port McNeill	1994		
<i>Growth And Yield Plots (1993)</i>	Port McNeill	1993		
<i>Growth And Yield Plots (1992)</i>	Port McNeill	1992		
<i>Growth And Yield Plots (1991)</i>	Port McNeill Jeune Landing	1991		
<i>Growth And Yield Plots (1989)</i>	Port McNeill Jeune Landig	1989		
<i>Growth And Yield Plots (1988)</i>	Port McNeill	1988		
<i>Site Index Species Conversion Surveys (1997)</i>	Port McNeill Holberg Jeune Landing	1997		

Appendix XIX



Doman Forest Products Limited Western Pulp Inc. Western Forest Products Limited Doman – Western Lumber Ltd.

Operating Statistics

Productive Forest Land Managed	885 000 ha
Operable Forest Land	550 000 ha
Forest Tenures	3 Tree Farm Licences 7 Forest Licences 5 Managed Forests 127 Timber Licences
Logging Operations	30
Employees and Contractors	4 200 people (est)
Annual Timber Harvest	4 200 000 m ³
Annual Timber Purchase	800 000 m ³
Mills	2 Pulp Mills – 1 Kraft, 1 Sulphite 9 Saw Mills 1 Value Added Plant 1 Log Merchandiser
Products: Lumber and Solid Wood	800 000 000 board feet
Pulp	400 000 tonnes
Annual Product Sales	over \$800 million
Annual Roads Construction	335 km
Annual Roads Maintenance	2 100 km
Annual Logging	5 500 ha
Annual Planting	5 100 ha
Annual Natural Regeneration	400 ha
Annual Number of Seedlings Planted	5 000 000 trees
Average Survival of 3-year Old Plantations	90 %
Annual Site Preparation	700 ha
Annual Brushing and Weeding	2 300 ha
Annual Juvenile Spacing	1 600 ha
Annual Pruning	1 000 ha
Annual Fertilization	3 000 ha
Annual visitors to Forest Lands	over 300 000 visitors
Annual Salmon Enhancement Production (4 hatcheries)	750 000 fry
Recreation Sites and Trails	45
Forest Enhancement Person-Days	over 40 000

Appendix XX

Summary of Obligations
Management Plan Commitments and Issues*
Status Report

Commitment	Timeline	Status
Refine Site Index/Ecosite Relationship	Prior to MP 9 draft	A project was completed by J.S.Thrower & Associates that confirmed relationship.
Confirm yield curve and stand density relationship	May 31, 1996 for review. Next MP for changes.	A project has been undertaken with Olympic Resources to utilize PSP data to project inventory data through time. The results will be available in 1999.
Address Inventory gaps	Prior to preparation of 20 year plan (Dec 96 Target).	Recreation – Opportunity mapping done and inputted into GIS. Terrain/Silviculture – Completed and inputted into GIS. Wildlife – 11 or 14 Deer Winter Ranges “grandfathered” as UWRs. Spatial habitat project to be completed in early 1999. Stream Classification – RIC inventory in progress to supplement operational work. Archaeological Sites – Captured from HCB records. Biodiversity – D.Blood report being supplemented by ecosystem and forest cover mapping. TRP – Process underway to use ATLAS for spatial mapping for MP 9.
Deciduous stands will be harvested and replaced with conifers through the rotation as opportunities arise.	No date.	Harvesting of deciduous stands was included as part of the long term harvesting plan in MP 8. However, such stand conversion was viewed as occurring concurrently with the harvest of adjacent second growth coniferous stands. WFP has not yet reached this point in time and very little harvesting is occurring in these younger stands. We have invited deciduous contractors to bid on stand conversion.
Accounting for non-recoverable losses.	During term of MP.	Pilot windthrow project is completed. Data being analysed for expanded program.
Feasibility study of commercial thinning	During term of MP.	A series of CT trials have been completed using different systems (cable, CTL and hoe-forwarding. Results are not encouraging for both economic and forest health reasons. Projects have been uneconomic and level of damage in hemlock stands has been unacceptable.

* Outstanding commitments and issues that were contained in the implementation section of the TFL 6 AAC rationale, the approval letter and in the AAC rationale report for management Plan 8.

Appendix XXI

Saanich Forestry Centre
Seedling Production Report

Species	Size	Seedlings Produced					
		Surplus	Fall	Spring	Total by Size	Total by Species	%
Cw	313B	0	0	552,480	552,480		
Cw	410A	0	0	244,020	244,020		
Cw	415C	7,690	1,182	720,170	729,042		
Cw	615A	0	800	25,615	26,415	1,551,957	50.25
Dr	415B	0	0	23,068	23,068	23,068	0.75
Ds	415B	0	0	38,160	38,160	38,160	1.24
Fc	313B	0	0	50,020	50,020		
Fc	410A	0	35,280	8,420	43,700		
Fc	415C	0	273	486,533	486,806		
Fc	615A	0	0	21,924	21,924	602,450	19.51
Hm	410A	0	0	18,370	18,370	18,370	0.59
Hw	313B	89,980	0	192,840	282,820		
Hw	415C	34,900	500	258,130	293,530	576,350	18.66
Pw	415B	0	0	5,296	5,296	5,296	0.17
Ss	313B	0	0	72,640	72,640		
Ss	410A	0	0	17,410	17,410		
Ss	415C	0	480	92,510	92,990	183,040	5.93
Sx	415C	6,770	0	41,570	48,340	48,340	1.57
Sxs	415C	0	12,400	3,030	15,430	15,430	0.50
Yc	410A	0	0	25,100	25,100	25,100	0.81
Misc.	313B	0	40	0	40		
Misc.	410A	0	100	0	100		
Misc.	415C	0	480	0	480	620	0.02
Total		139,340	51,535	2,897,306	3,088,181	3,088,181	100.00

Appendix XXII

Saanich Forestry Centre and Lost Lake Seed Orchard
Seed Production Report

Seedlot	Species	Orchard	Zone	BVvol60 ¹ (%)	Elevation (m)	Latitude	Longitude	Volume (hL)	Seed Weight (Kg)	Seedlings (estimated)	Comments
60350	Fdc	166	M/GL	17.5	304	49 13	123 35	1.485	0.578	28,113	Elite Crosses (FRBC Funded)
60351	Hw	126	M	15.7	121	50 02	125 52	0.364	0.252	61,039	Elite Crosses (FRBC Funded)
60352	Hw	126	M	7.9	95	50 30	126 53	1.928	1.334	323,304	High BV Bulk collections
60353	Cw	128	M	N/A	168	50 45	127 35	5.026	3.129	616,172	22% midge - 1/2 crop extracted MoF
60355	Cw	128	M	N/A	168	50 45	127 35	5.026	3.129	616,172	22% midge - 1/2 crop extracted YPP
60354	Cw	155	M	N/A	252	52 42	131 39	5.487	4.029	793,403	8% midge - 1/2 crop extracted MoF
60356	Cw	155	M	N/A	252	52 42	131 39	5.487	4.029	793,403	8% midge - 1/2 crop extracted YPP
Total:									16.48	3,231,606	

¹ Expected gain in volume over wild seedlots at rotation.